US 20250286670A1

(19) United States

(12) Patent Application Publication

(10) Pub. No.: US 2025/0286670 A1

WU et al.

(43) Pub. Date: Sep. 11, 2025

(54) **WIDEBAND OPERATION WITH MULTIPLE RADIO FREQUENCY (RF) CHAINS**

(71) Applicant: **QUALCOMM Incorporated**, San Diego, CA (US)

(72) Inventors: **Shijun WU**, San Diego, CA (US); **Kangqi LIU**, San Diego, CA (US); **Weimin DUAN**, San Diego, CA (US)

(21) Appl. No.: **18/598,945**

(22) Filed: **Mar. 7, 2024**

Publication Classification

(51) **Int. Cl.**

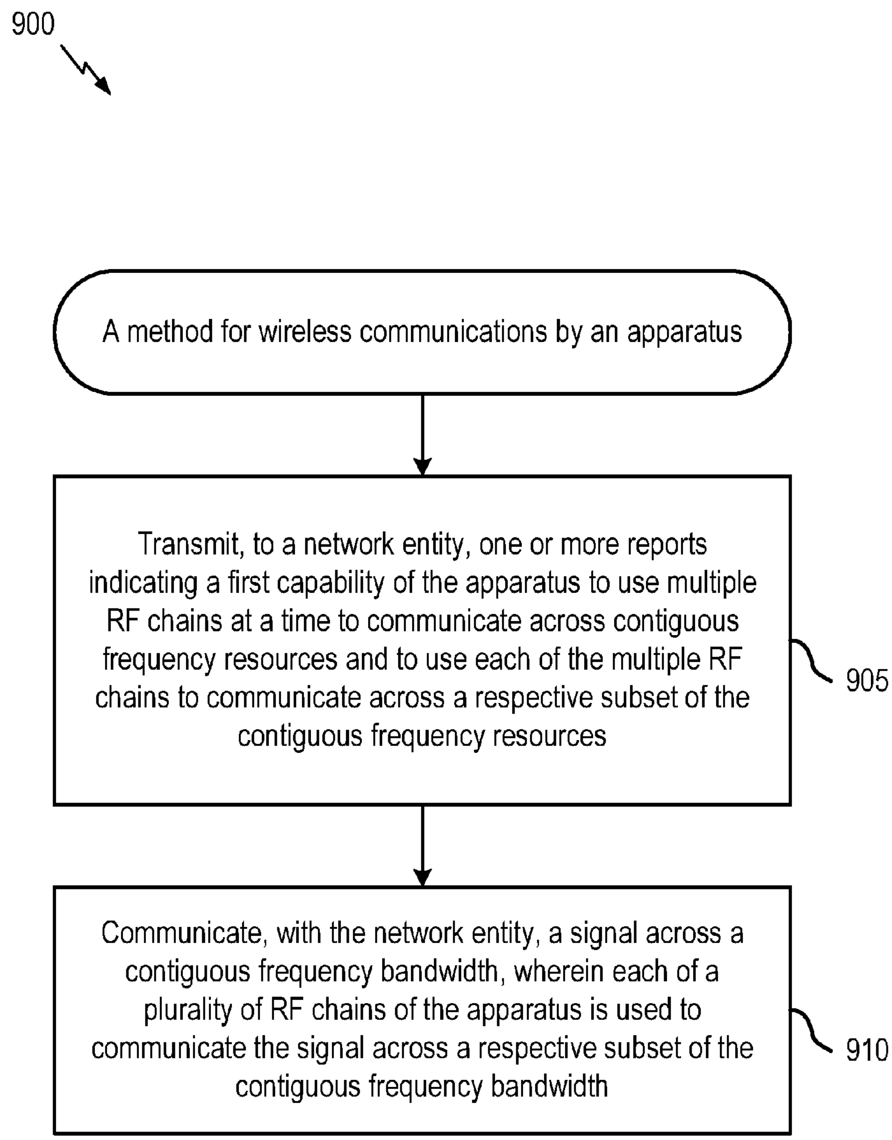
H04L 5/00	(2006.01)
H04B 7/0413	(2017.01)
H04L 27/26	(2006.01)

(52) **U.S. Cl.**

CPC ***H04L 5/0005*** (2013.01); ***H04B 7/0413*** (2013.01); ***H04L 5/0039*** (2013.01); ***H04L 27/26524*** (2021.01)

(57) **ABSTRACT**

Certain aspects of the present disclosure provide techniques for using multiple radio frequency (RF) chains at a time for wireless communications. A method generally includes transmitting, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and communicating, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth.



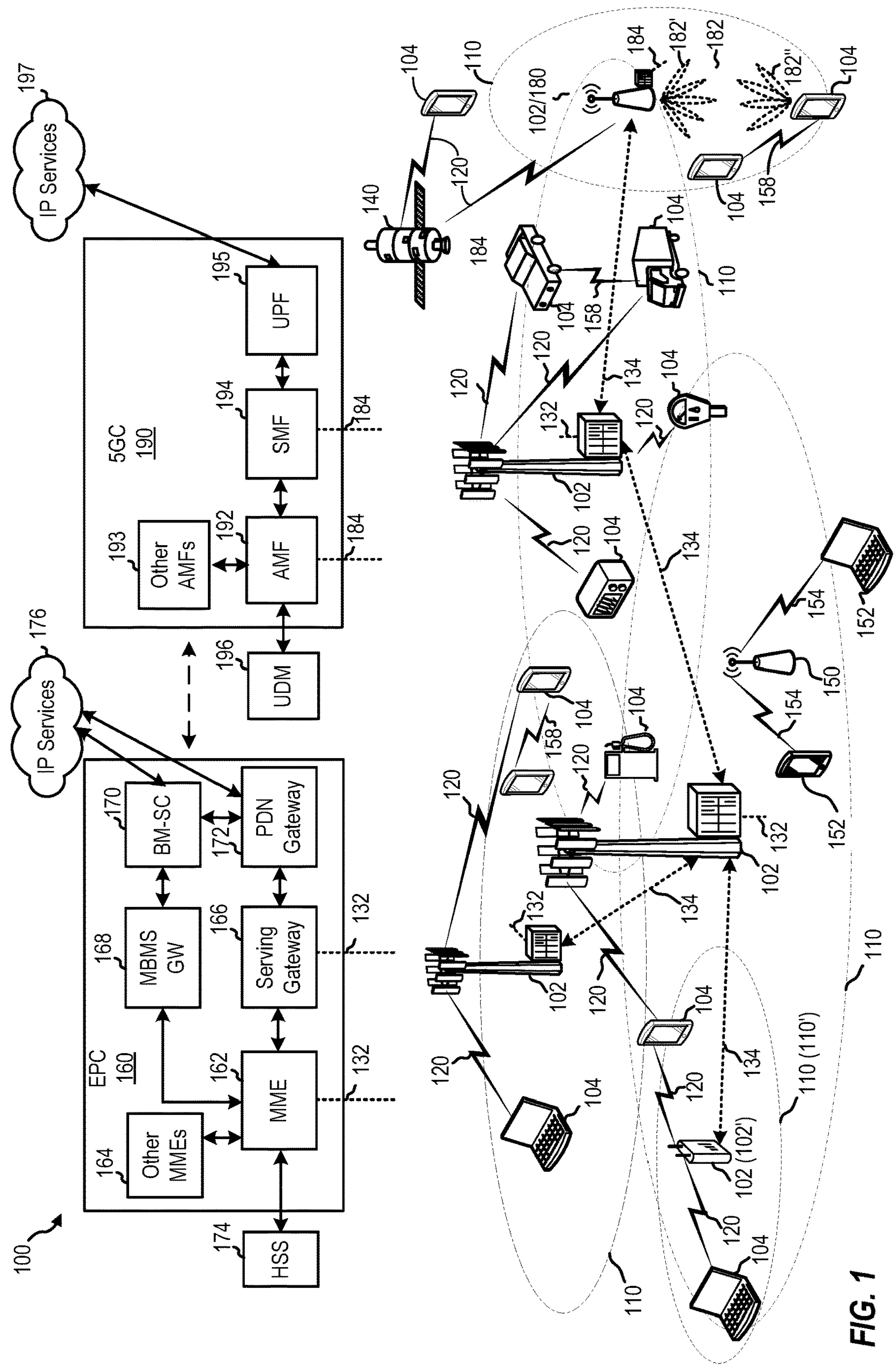

100
EPC
160
HSS
174
Other MMEs
164
MME
162
MBMS GW
168
Serving Gateway
166
BM-SC
170
PDN Gateway
172
IP Services
176
5GC
190
Other AMFs
193
AMF
192
SMF
194
UPF
195
UDM
196
IP Services
197
132
184
102
104
110
110 (110')
102 (102')
120
134
140
150
152
154
158
102/180
182'
182"
182

FIG. 1

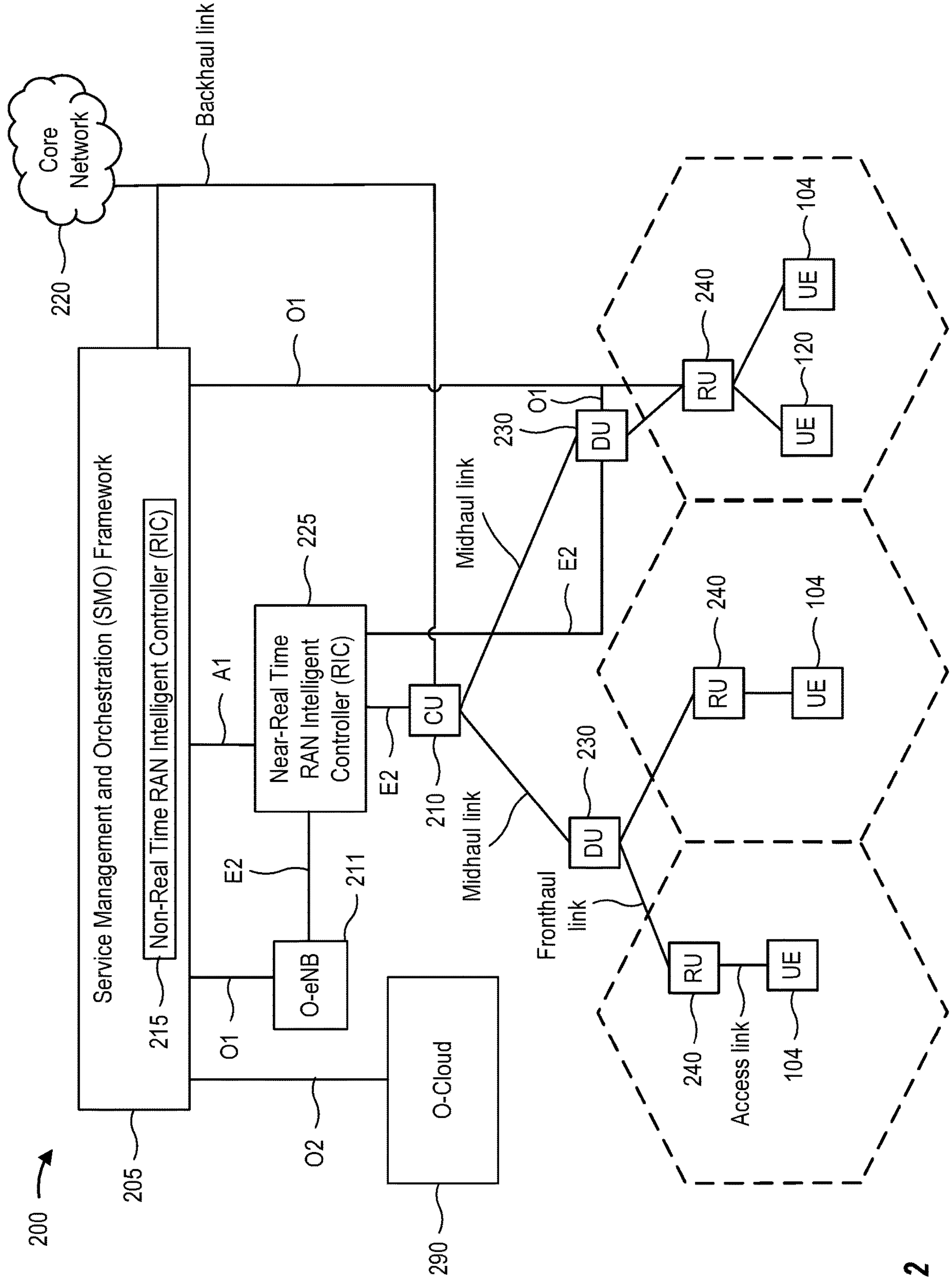

200
Core Network
220
Backhaul link
Service Management and Orchestration (SMO) Framework
Non-Real Time RAN Intelligent Controller (RIC)
215
205
O1
O1
A1
Near-Real Time RAN Intelligent Controller (RIC)
225
E2
E2
E2
O-eNB
211
O1
O2
O-Cloud
290
CU
210
Midhaul link
Midhaul link
DU
230
DU
230
O1
Fronthaul link
RU
240
RU
240
RU
240
UE
104
UE
120
UE
104
UE
104
Access link

FIG. 2

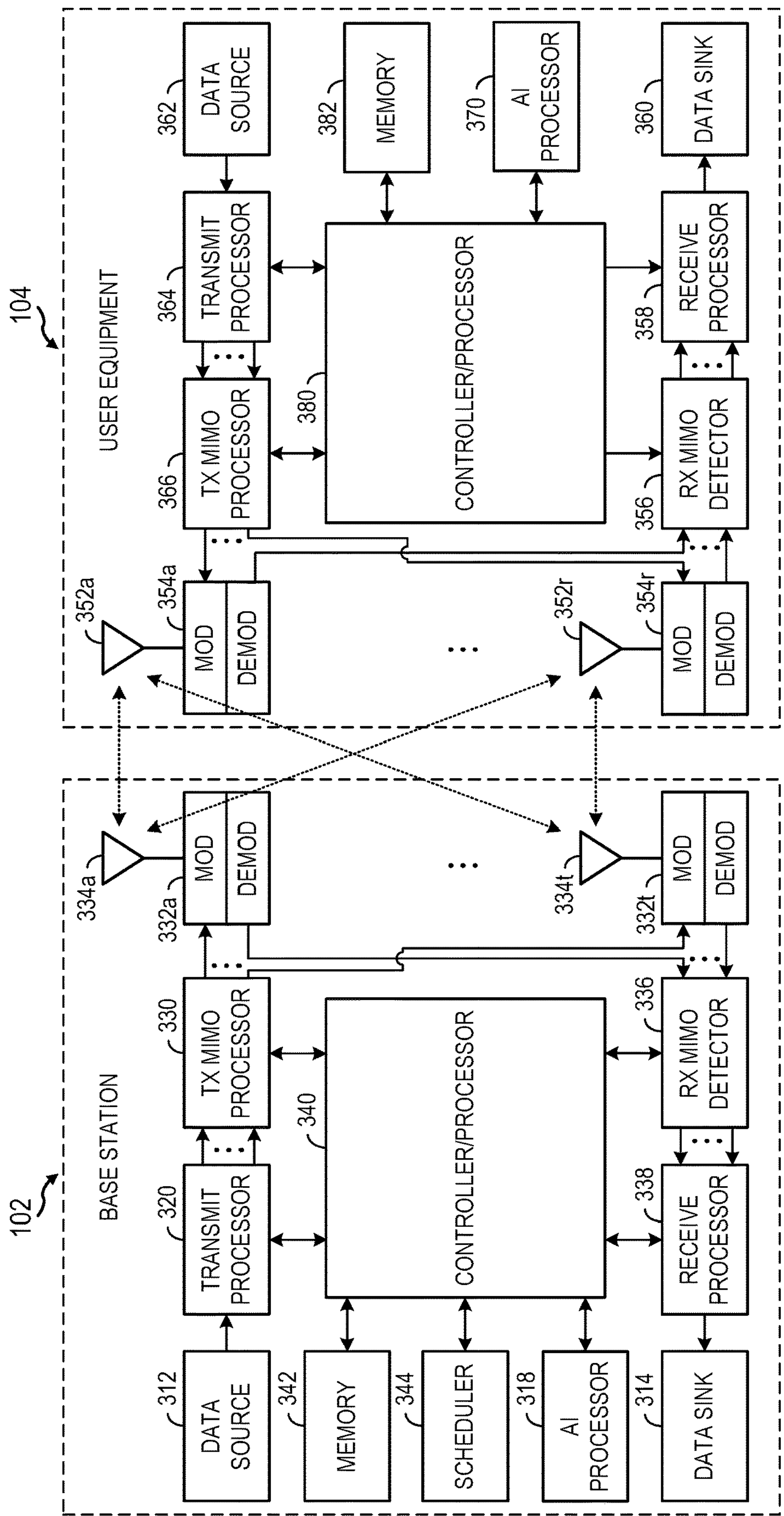

104
USER EQUIPMENT
362
DATA SOURCE
364
TRANSMIT PROCESSOR
366
TX MIMO PROCESSOR
352a
354a
MOD
DEMOD
382
MEMORY
380
CONTROLLER/PROCESSOR
370
AI PROCESSOR
360
DATA SINK
358
RECEIVE PROCESSOR
356
RX MIMO DETECTOR
352r
354r
MOD
DEMOD
102
BASE STATION
312
DATA SOURCE
320
TRANSMIT PROCESSOR
330
TX MIMO PROCESSOR
332a
334a
MOD
DEMOD
342
MEMORY
344
SCHEDULER
318
AI PROCESSOR
340
CONTROLLER/PROCESSOR
314
DATA SINK
338
RECEIVE PROCESSOR
336
RX MIMO DETECTOR
332t
334t
MOD
DEMOD

FIG. 3

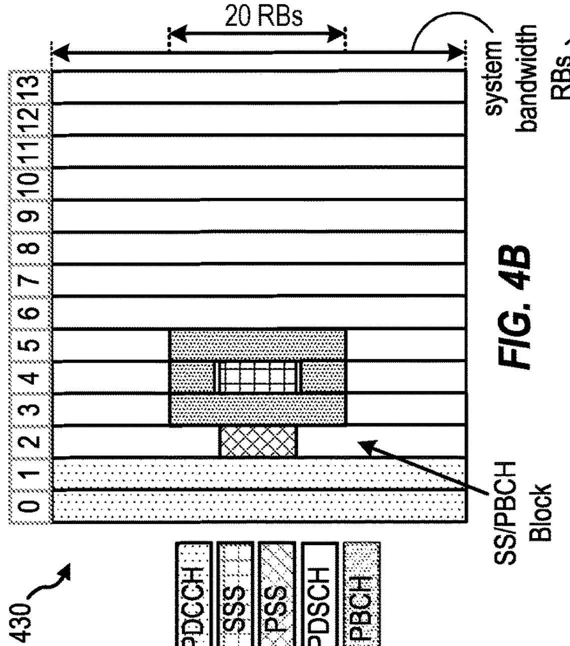


FIG. 4B

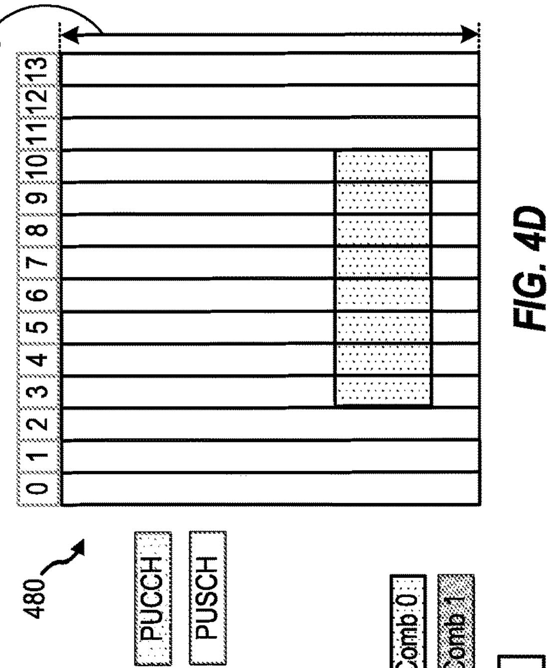


FIG. 4D

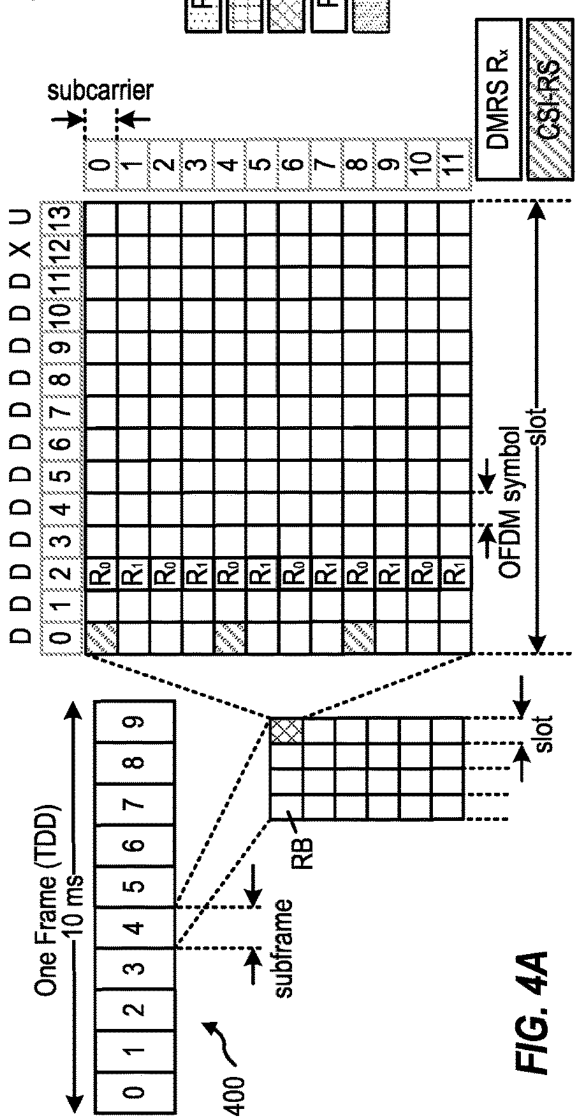


FIG. 4A

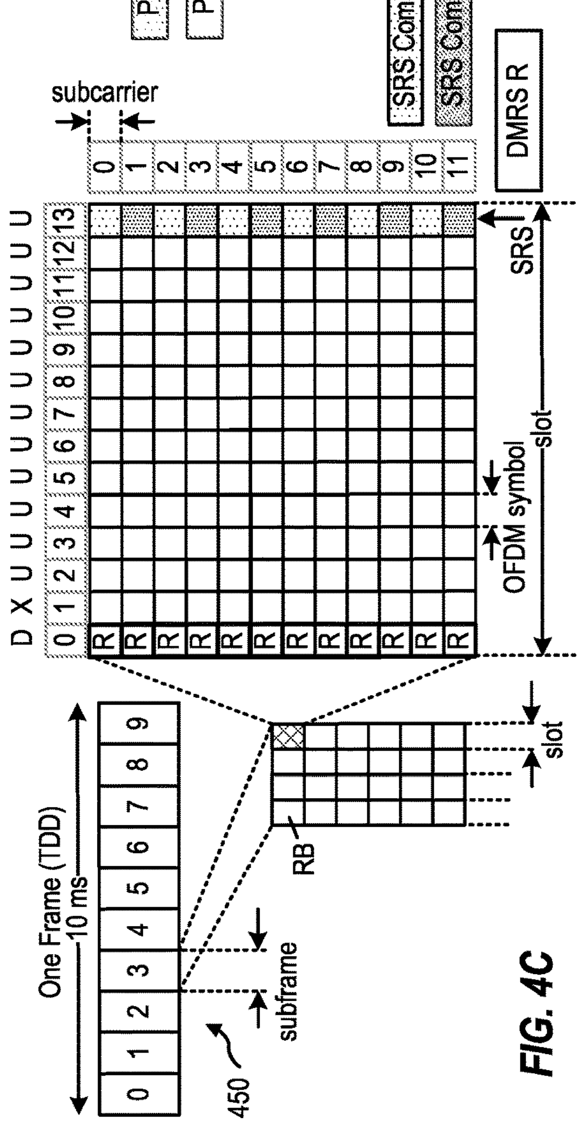


FIG. 4C

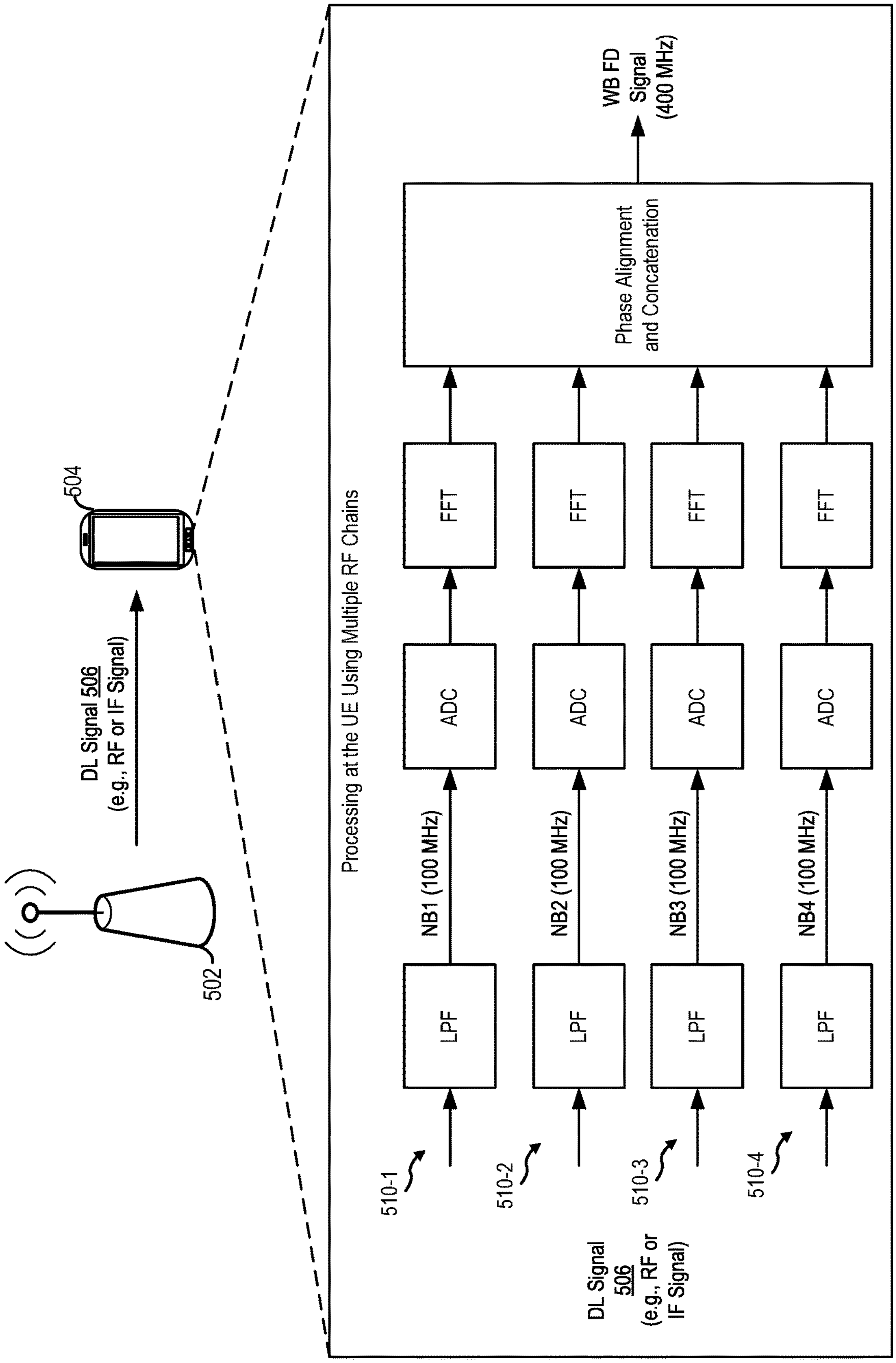
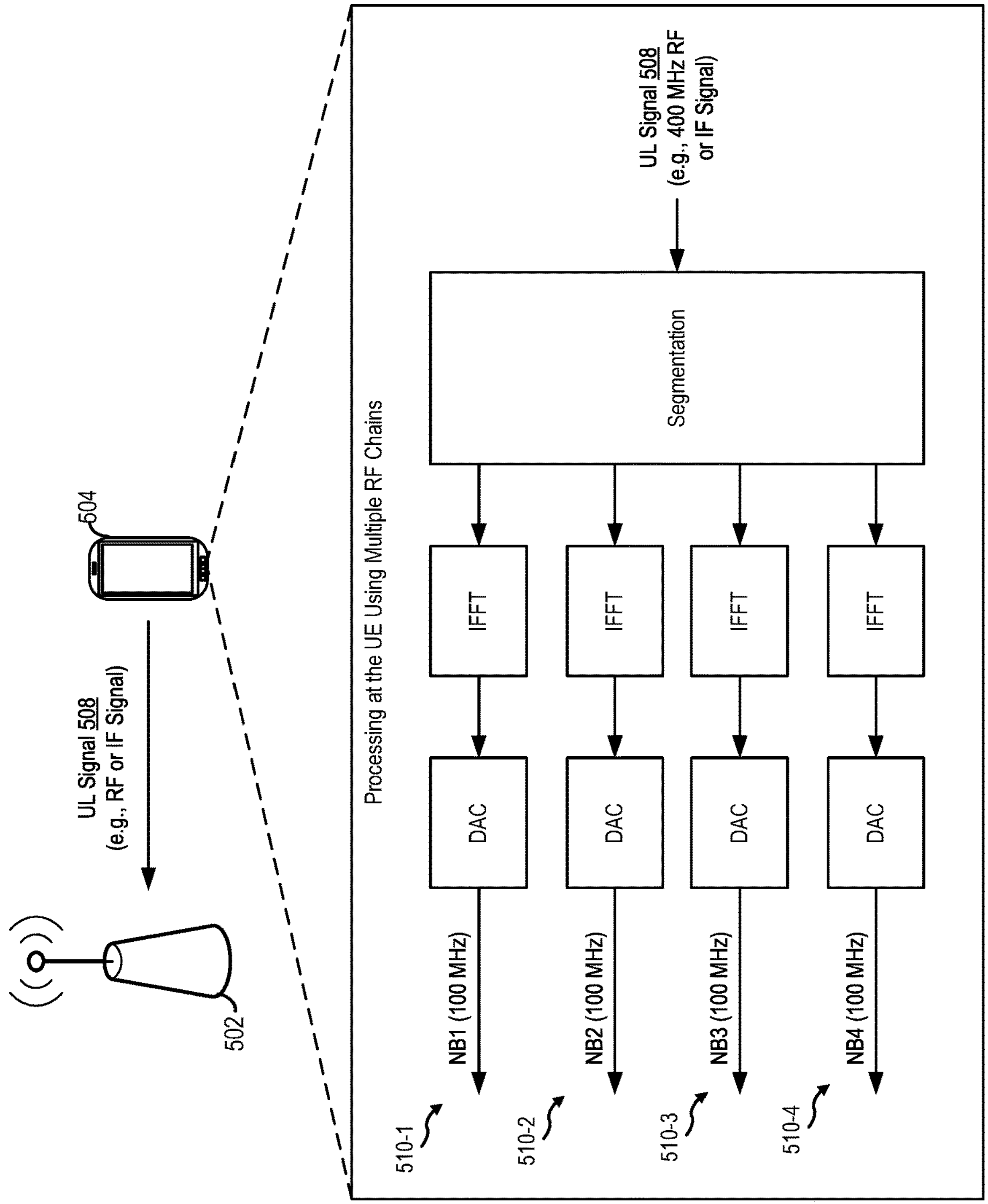

502
DL Signal 506
(e.g., RF or IF Signal)
504
Processing at the UE Using Multiple RF Chains
DL Signal 506 (e.g., RF or IF Signal)
510-1
510-2
510-3
510-4
LPF
LPF
LPF
LPF
NB1 (100 MHz)
NB2 (100 MHz)
NB3 (100 MHz)
NB4 (100 MHz)
ADC
ADC
ADC
ADC
FFT
FFT
FFT
FFT
Phase Alignment and Concatenation
WB FD Signal (400 MHz)

FIG. 5A



UL Signal 508
(e.g., RF or IF Signal)
502
504
Processing at the UE Using Multiple RF Chains
UL Signal 508
(e.g., 400 MHz RF
or IF Signal)
Segmentation
IFFT
IFFT
IFFT
IFFT
DAC
DAC
DAC
DAC
NB1 (100 MHz)
NB2 (100 MHz)
NB3 (100 MHz)
NB4 (100 MHz)
510-1
510-2
510-3
510-4

FIG. 5B

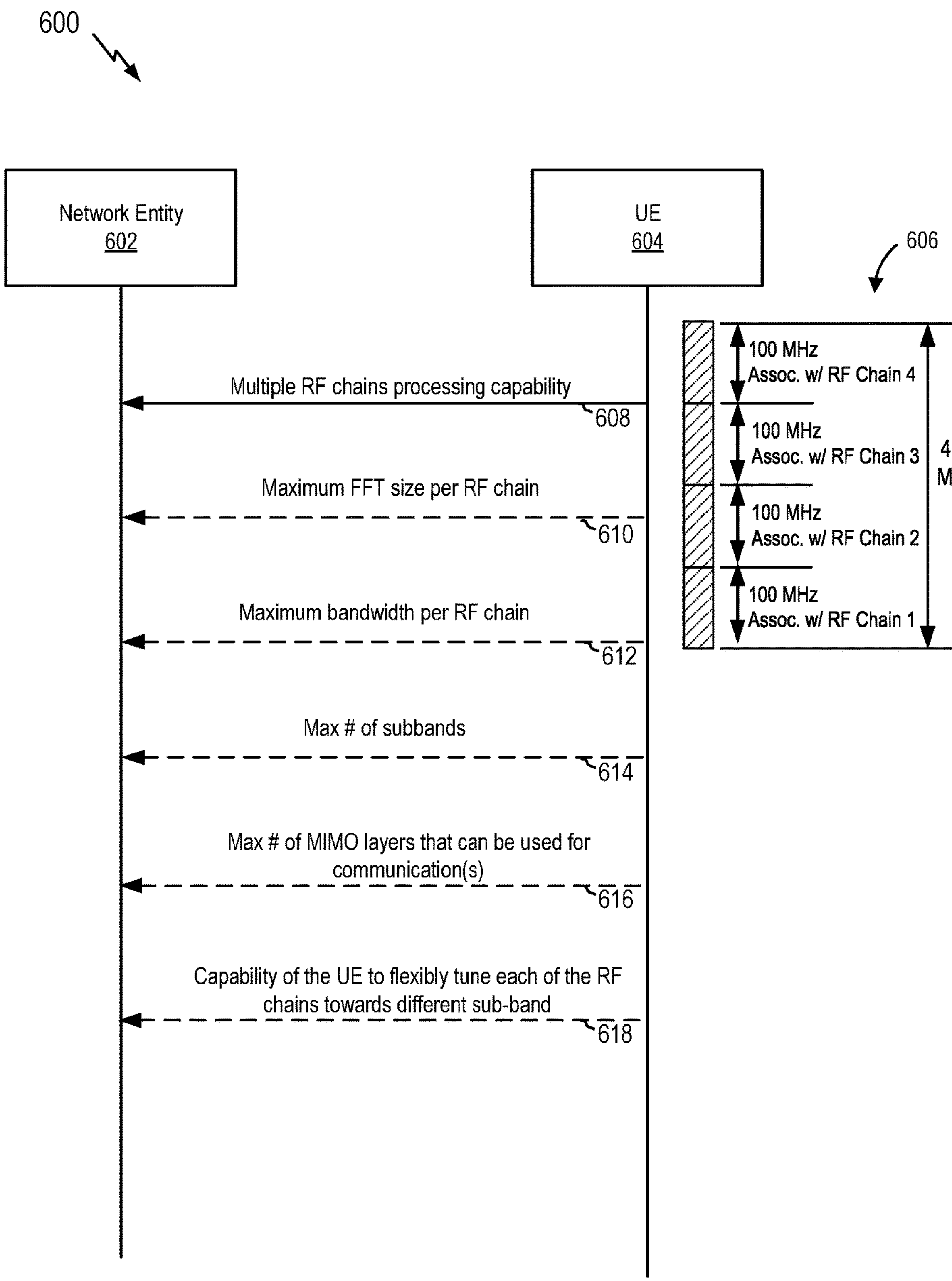

600
Network Entity
602
UE
604
606
Multiple RF chains processing capability
608
Maximum FFT size per RF chain
610
Maximum bandwidth per RF chain
612
Max # of subbands
614
Max # of MIMO layers that can be used for communication(s)
616
Capability of the UE to flexibly tune each of the RF chains towards different sub-band
618
100 MHz Assoc. w/ RF Chain 4
100 MHz Assoc. w/ RF Chain 3
100 MHz Assoc. w/ RF Chain 2
100 MHz Assoc. w/ RF Chain 1
400 MHz

FIG. 6

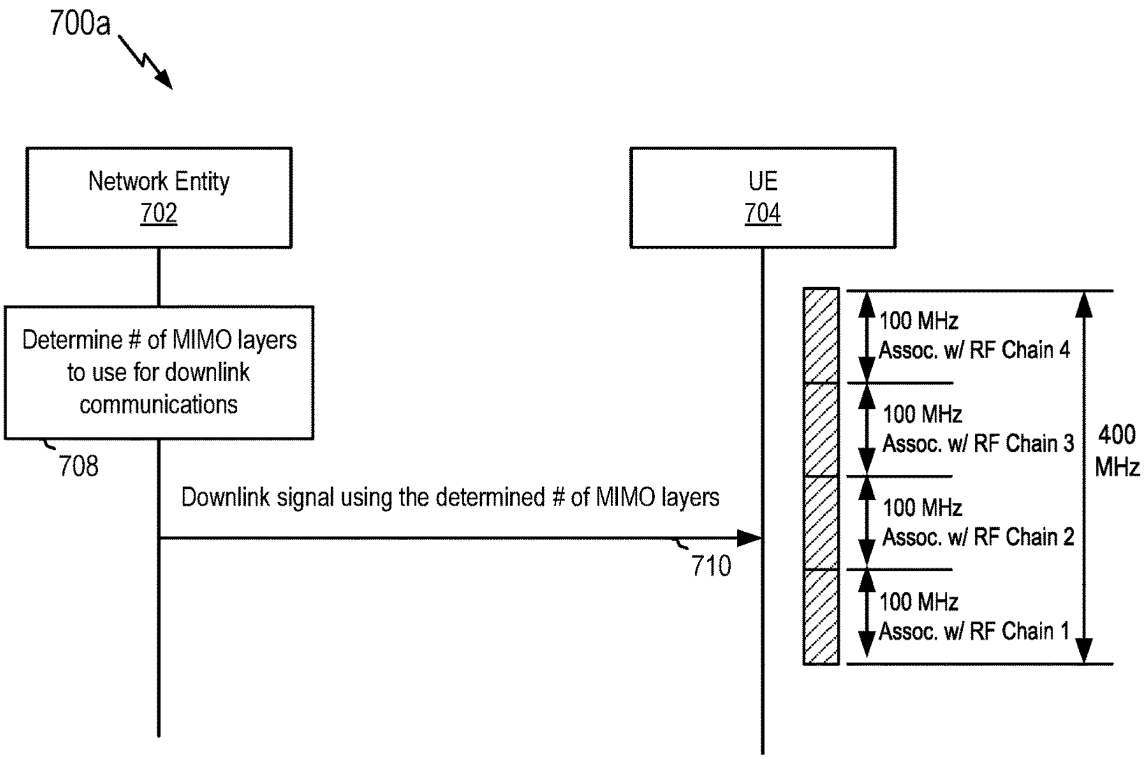

700a
Network Entity
702
UE
704
Determine # of MIMO layers to use for downlink communications
708
Downlink signal using the determined # of MIMO layers
710
100 MHz
Assoc. w/ RF Chain 4
100 MHz
Assoc. w/ RF Chain 3
100 MHz
Assoc. w/ RF Chain 2
100 MHz
Assoc. w/ RF Chain 1
400 MHz

FIG. 7A

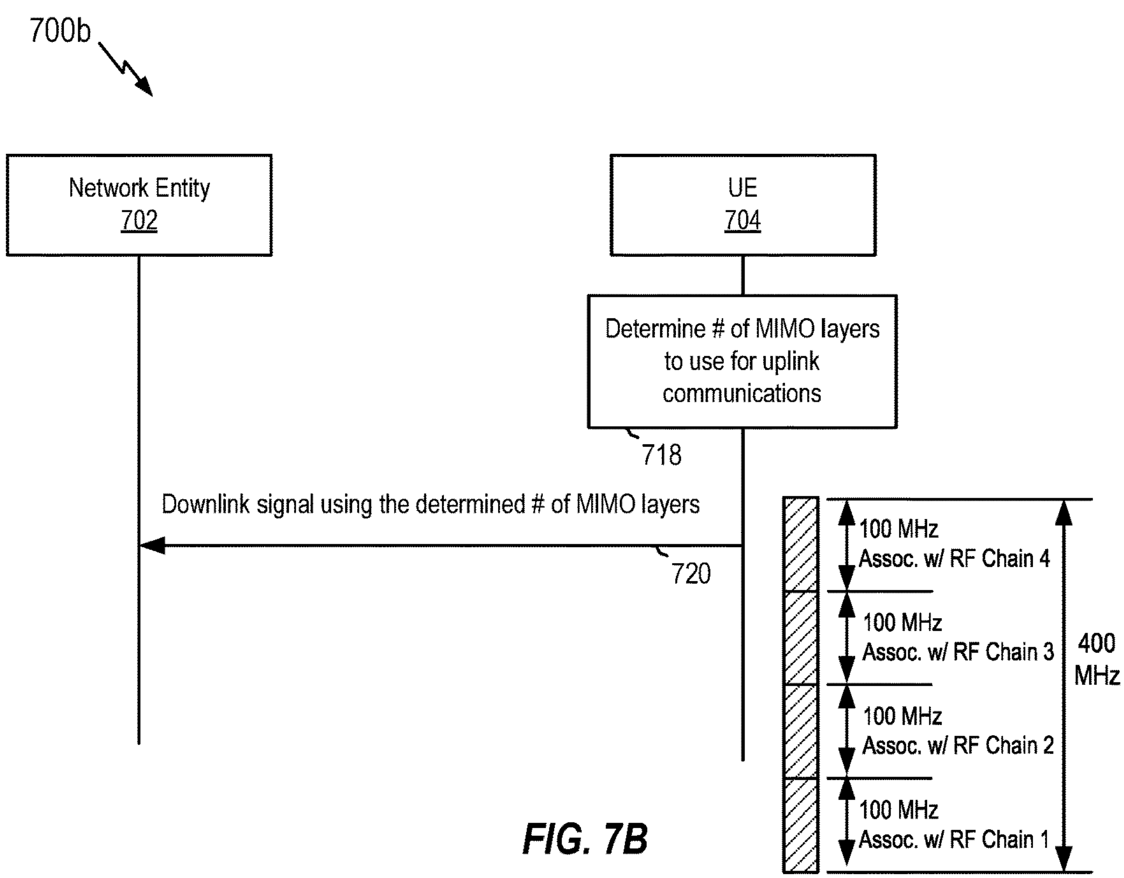

700b
Network Entity
702
UE
704
Determine # of MIMO layers to use for uplink communications
718
Downlink signal using the determined # of MIMO layers
720
100 MHz
Assoc. w/ RF Chain 4
100 MHz
Assoc. w/ RF Chain 3
100 MHz
Assoc. w/ RF Chain 2
100 MHz
Assoc. w/ RF Chain 1
400 MHz

FIG. 7B

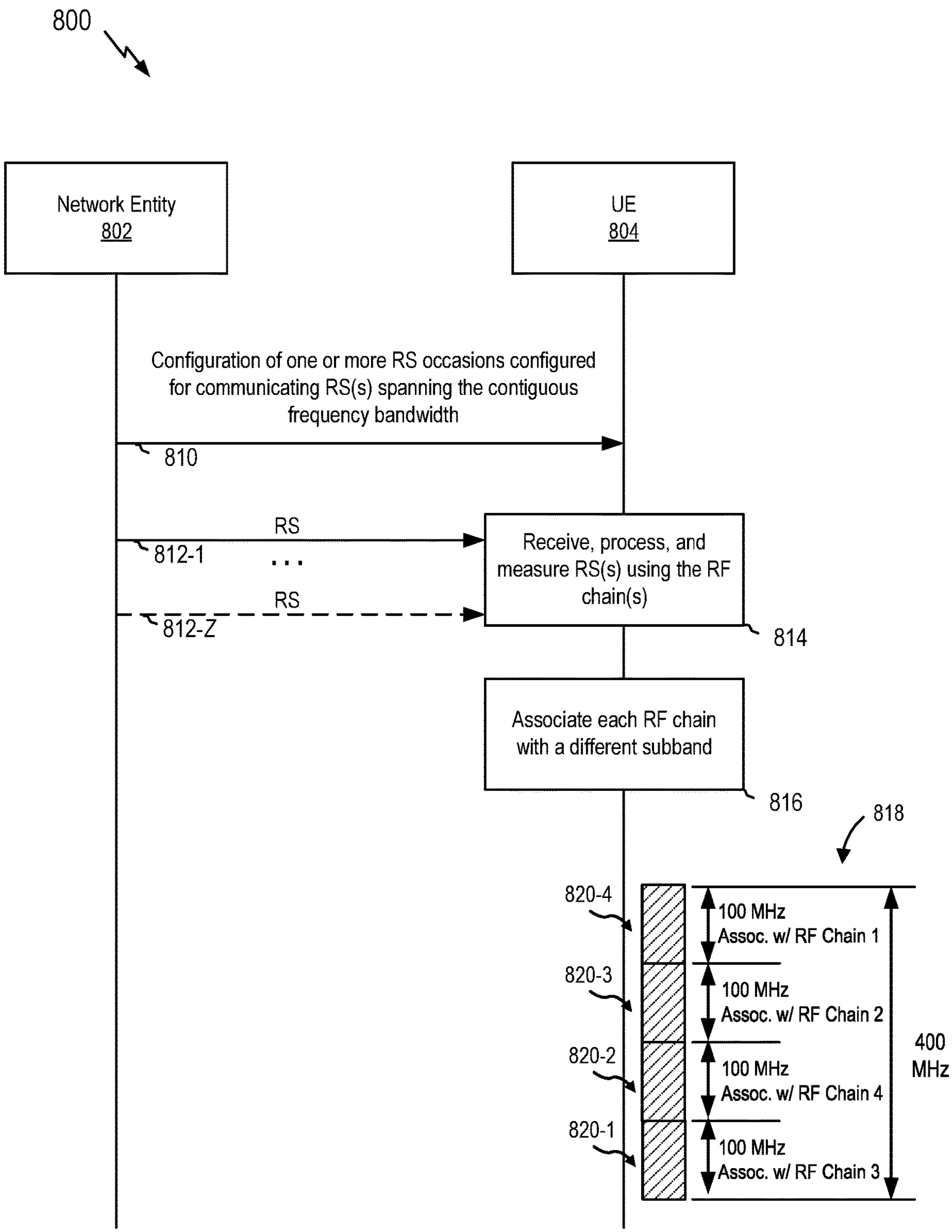
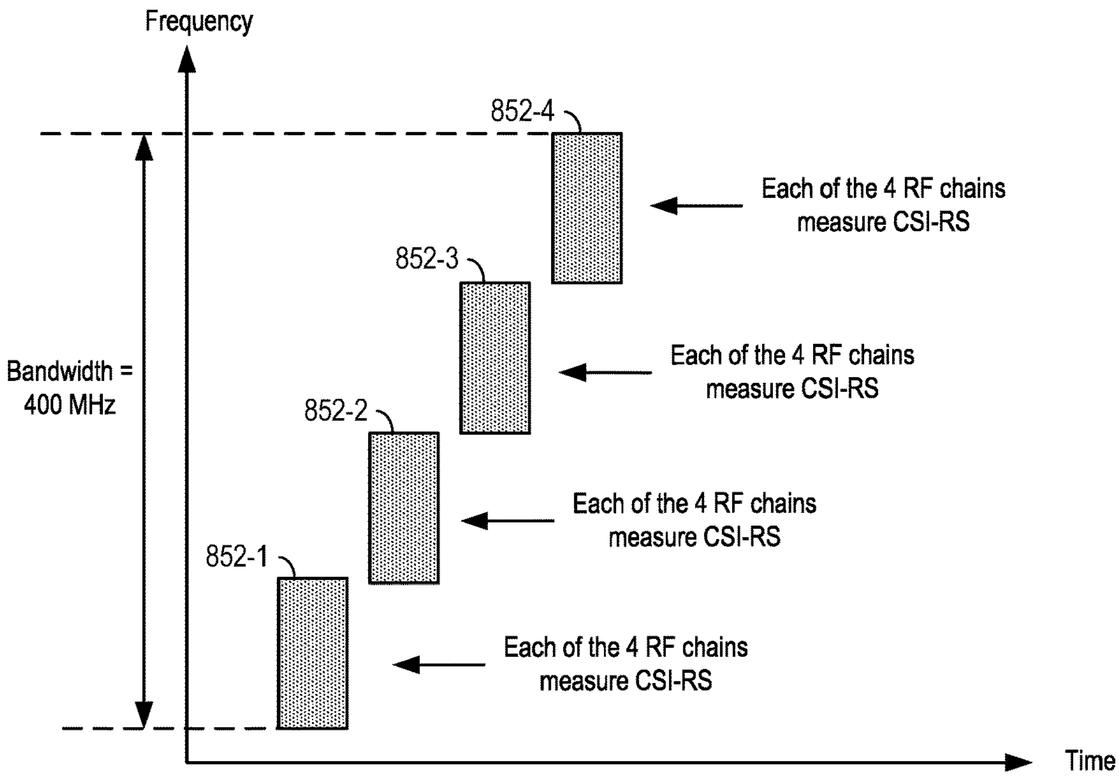
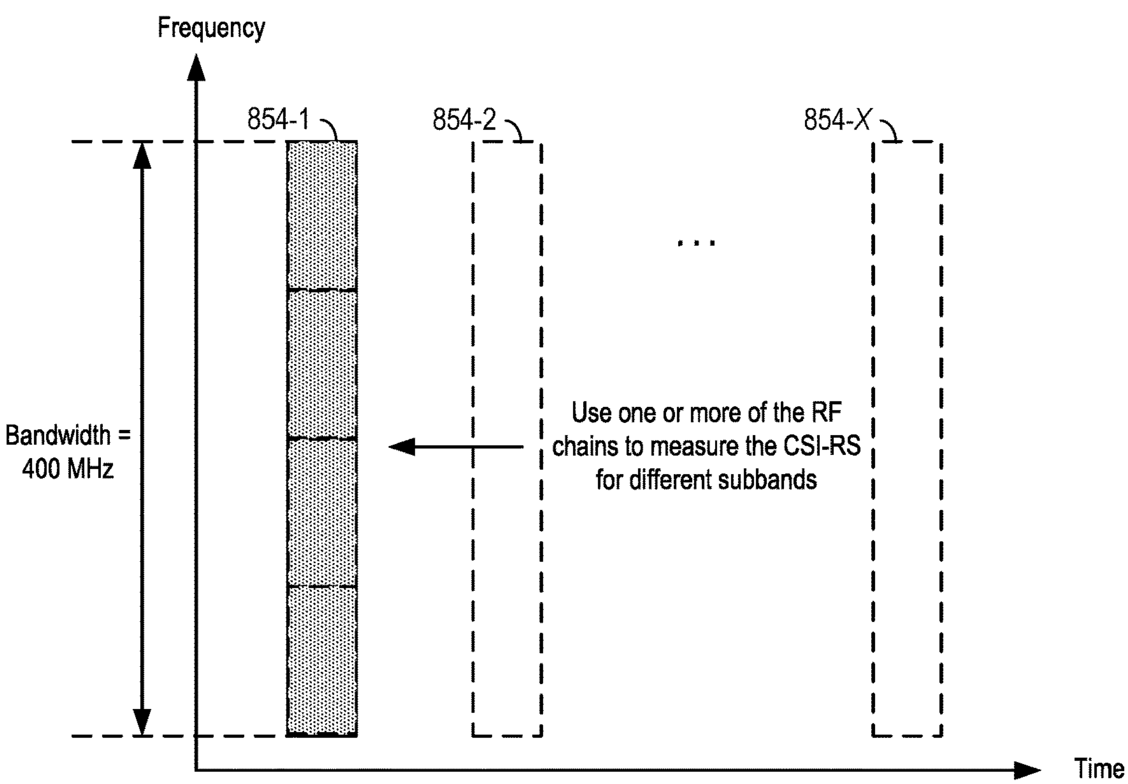

800
Network Entity
802
UE
804
Configuration of one or more RS occasions configured for communicating RS(s) spanning the contiguous frequency bandwidth
810
RS
812-1
RS
812-Z
Receive, process, and measure RS(s) using the RF chain(s)
814
Associate each RF chain with a different subband
816
818
820-4
100 MHz
Assoc. w/ RF Chain 1
820-3
100 MHz
Assoc. w/ RF Chain 2
820-2
100 MHz
Assoc. w/ RF Chain 4
820-1
100 MHz
Assoc. w/ RF Chain 3
400
MHz

FIG. 8A



Frequency
852-4
Each of the 4 RF chains measure CSI-RS
852-3
Each of the 4 RF chains measure CSI-RS
Bandwidth = 400 MHz
852-2
Each of the 4 RF chains measure CSI-RS
852-1
Each of the 4 RF chains measure CSI-RS
Time

FIG. 8B



Frequency
854-1
854-2
854-X
. . .
Bandwidth = 400 MHz
Use one or more of the RF chains to measure the CSI-RS for different subbands
Time

FIG. 8C

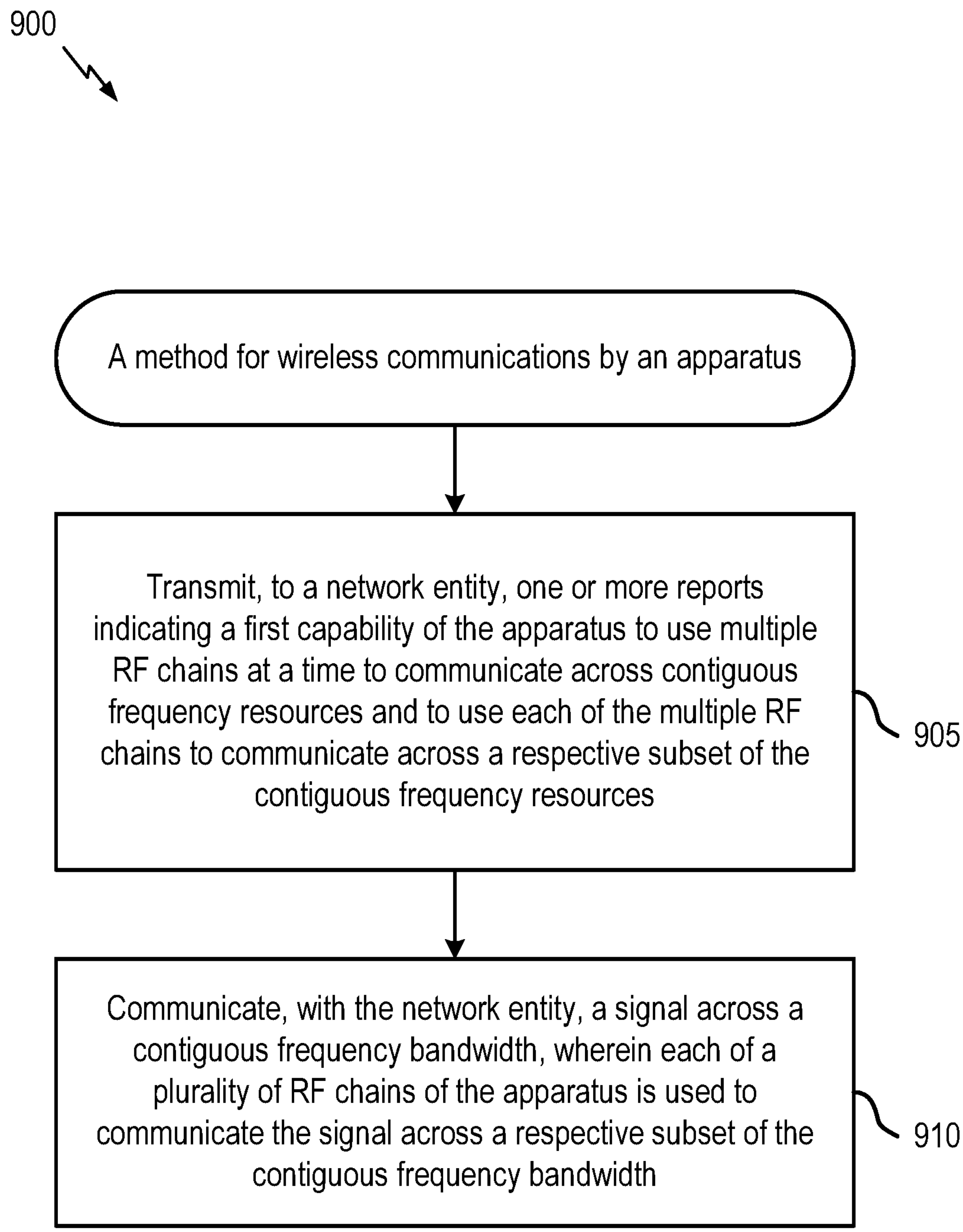

900
A method for wireless communications by an apparatus
Transmit, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources
905
Communicate, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth
910

FIG. 9

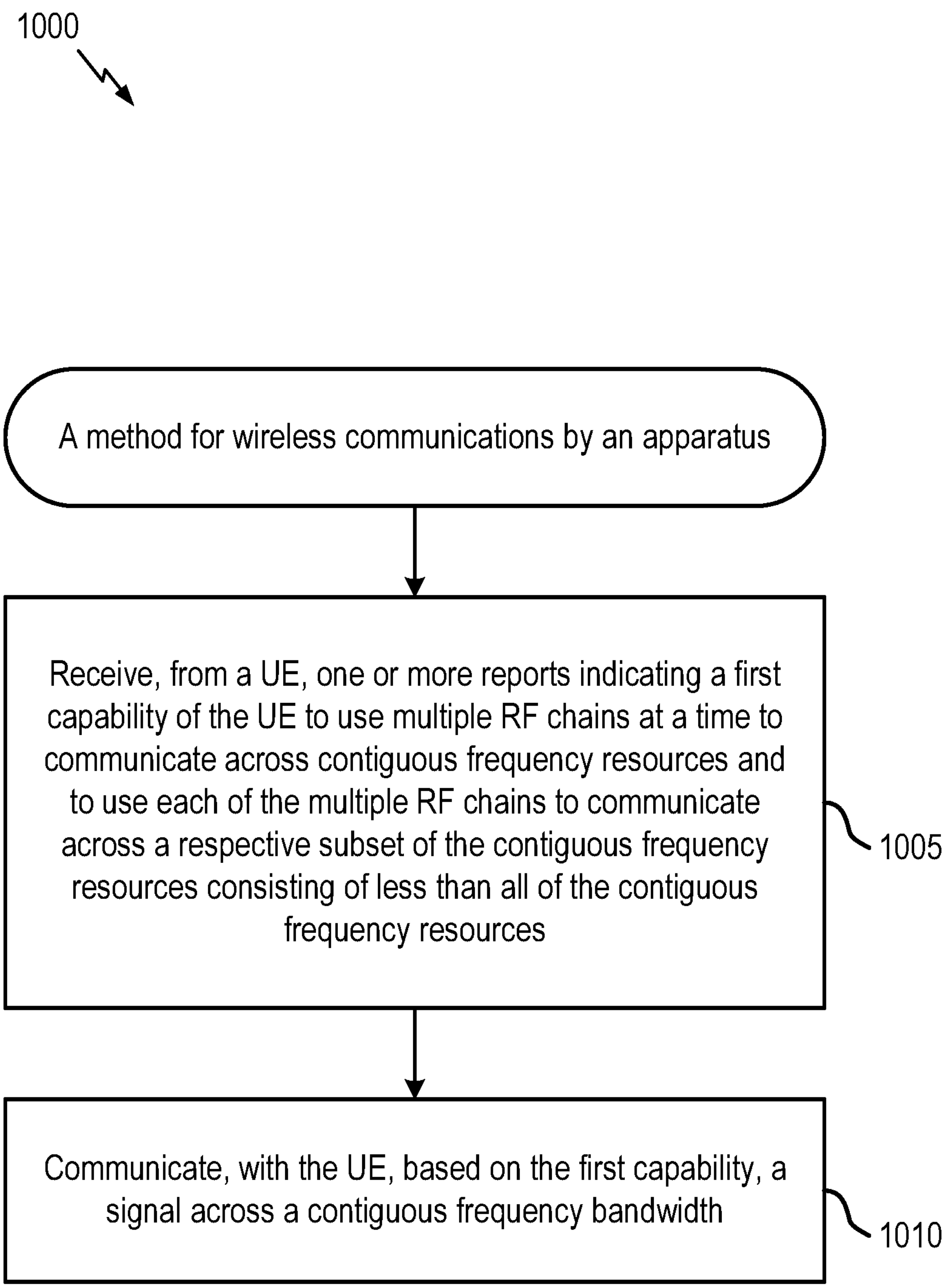

1000
A method for wireless communications by an apparatus
Receive, from a UE, one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources
1005
Communicate, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth
1010

FIG. 10

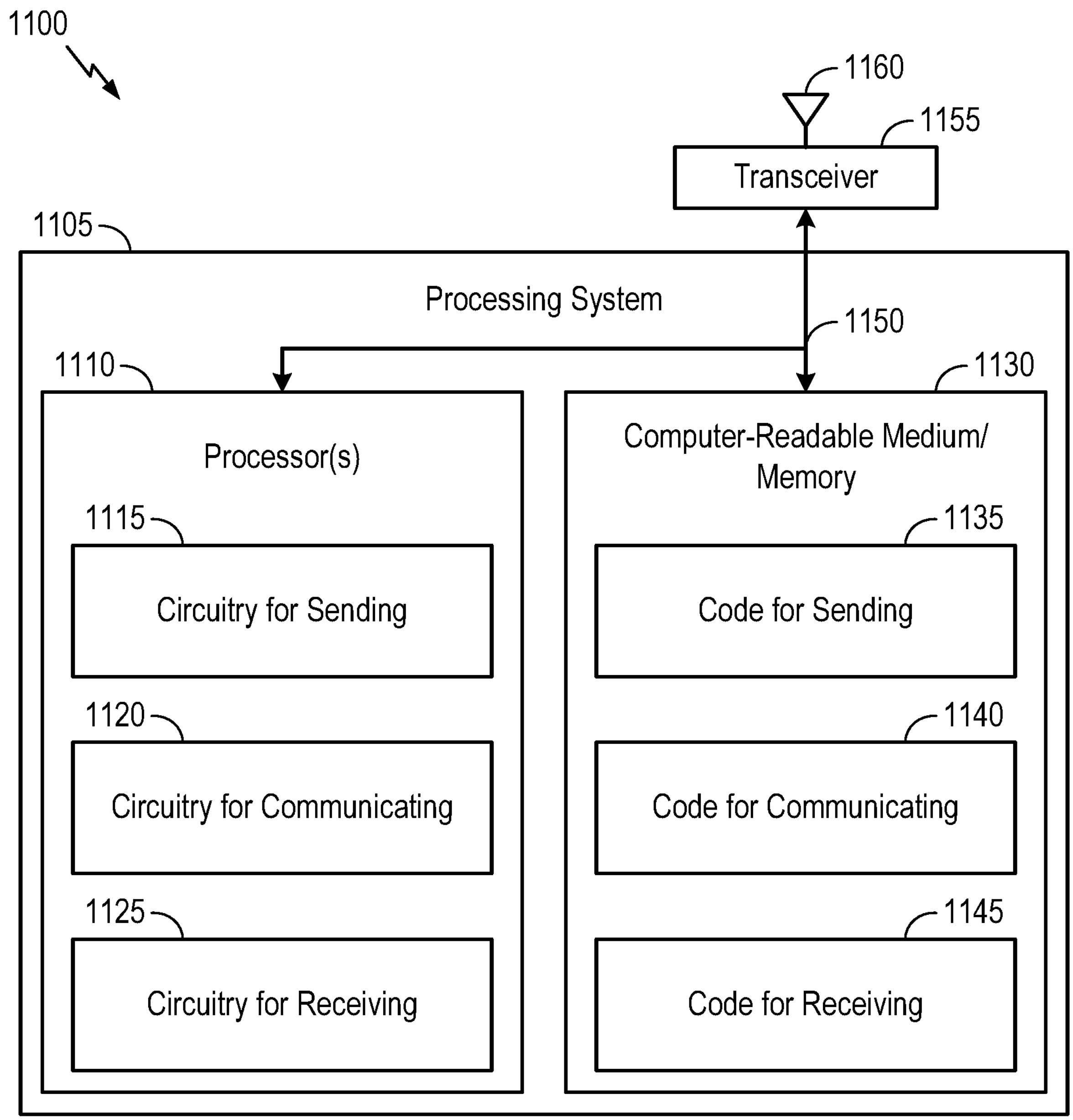

1100
1160
1155
Transceiver
1105
Processing System
1150
1110
1130
Processor(s)
Computer-Readable Medium/
Memory
1115
1135
Circuitry for Sending
Code for Sending
1120
1140
Circuitry for Communicating
Code for Communicating
1125
1145
Circuitry for Receiving
Code for Receiving

FIG. 11

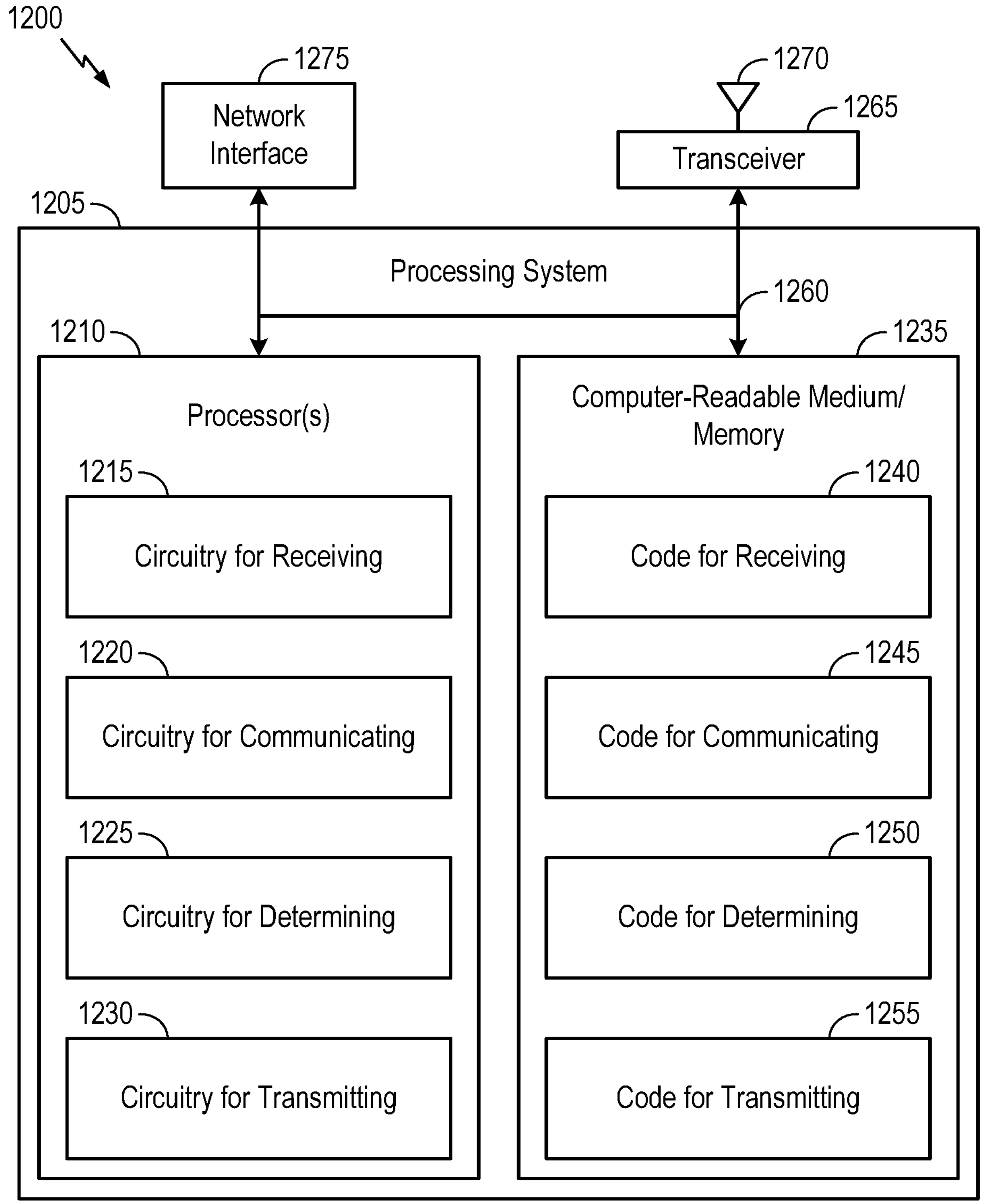

1200
1275
Network Interface
1270
1265
Transceiver
1205
Processing System
1260
1210
1235
Processor(s)
Computer-Readable Medium/ Memory
1215
Circuitry for Receiving
1240
Code for Receiving
1220
Circuitry for Communicating
1245
Code for Communicating
1225
Circuitry for Determining
1250
Code for Determining
1230
Circuitry for Transmitting
1255
Code for Transmitting

FIG.12

WIDEBAND OPERATION WITH MULTIPLE RADIO FREQUENCY (RF) CHAINS

INTRODUCTION

Field of the Disclosure

[0001] Aspects of the present disclosure relate to wireless communications, and more particularly, to techniques for multiple radio frequency (RF) chains processing.

Description of Related Art

[0002] Wireless communications systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, broadcasts, or other similar types of services. These wireless communications systems may employ multiple-access technologies capable of supporting communications with multiple users by sharing available wireless communications system resources with those users.

[0003] Although wireless communications systems have made great technological advancements over many years, challenges still exist. For example, complex and dynamic environments can still attenuate or block signals between wireless transmitters and wireless receivers. Accordingly, there is a continuous desire to improve the technical performance of wireless communications systems, including, for example: improving speed and data carrying capacity of communications, improving efficiency of the use of shared communications mediums, reducing power used by transmitters and receivers while performing communications, improving reliability of wireless communications, avoiding redundant transmissions and/or receptions and related processing, improving the coverage area of wireless communications, increasing the number and types of devices that can access wireless communications systems, increasing the ability for different types of devices to intercommunicate, increasing the number and type of wireless communications mediums available for use, and the like. Consequently, there exists a need for further improvements in wireless communications systems to overcome the aforementioned technical challenges and others.

SUMMARY

[0004] One aspect provides a method for wireless communications by an apparatus. The method includes transmitting, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple radio frequency (RF) chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and communicating, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth.

[0005] Another aspect provides one or more apparatuses configured for wireless communications. The one or more apparatuses include one or more memories and one or more processors, coupled to the one or more memories, configured to cause the one or more apparatuses to transmit, to a network entity, one or more reports indicating a first capability of the one or more apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and communicate, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the one or more apparatuses is used to communicate the signal across a respective subset of the contiguous frequency bandwidth. For example, performance of any of the above steps may be by only one apparatus or by multiple apparatuses in a distributed fashion.

[0006] Another aspect provides one or more apparatuses configured for wireless communications. The one or more apparatuses include means for transmitting, to a network entity, one or more reports indicating a first capability of the one or more apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and means for communicating, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the one or more apparatuses is used to communicate the signal across a respective subset of the contiguous frequency bandwidth. For example, performance of any of the above steps may be by only one apparatus or by multiple apparatuses in a distributed fashion.

[0007] Another aspect provides one or more non-transitory computer-readable media. The one or more non-transitory computer-readable media include executable instructions that, when executed by one or more processors of one or more apparatuses, cause the one or more apparatuses to transmit, to a network entity, one or more reports indicating a first capability of the one or more apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and communicate, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the one or more apparatuses is used to communicate the signal across a respective subset of the contiguous frequency bandwidth. For example, the instructions may be included in only one computer-readable medium or in a distributed fashion across multiple computer-readable media, such that instructions may be executed by only one processor or by multiple processors in a distributed fashion, such that each apparatus of the one or more apparatuses may include one processor or multiple processors, and/or such that performance may be by only one apparatus or in a distributed fashion across multiple apparatuses.

[0008] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum fast Fourier transform (FFT) size per RF chain supported by the one or more apparatuses.

[0009] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum bandwidth per RF chain supported by the one or more apparatuses.

[0010] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the one or more apparatuses.

[0011] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum number of multi-input-multiple-output (MIMO) layers supported by the one or more apparatuses when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

[0012] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, a maximum number of MIMO layers supported by the one or more apparatuses when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of: a maximum bandwidth; a subcarrier spacing (SCS) configured for communication between the network entity and the one or more apparatuses; a number of RF chains of the one or more apparatuses configured for communication between the network entity and the one or more apparatuses; a maximum FFT size per RF chain supported by the one or more apparatuses; a maximum bandwidth per RF chain supported by the one or more apparatuses; or a maximum number of subsets of contiguous frequency resources supported by the one or more apparatuses.

[0013] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the maximum bandwidth comprises a bandwidth configured for communication between the network entity and the one or more apparatuses.

[0014] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the maximum bandwidth comprises a nominal maximum system bandwidth of the network entity.

[0015] Some examples of the methods, apparatuses, and non-transitory computer-readable media described herein may further include operations, features, means, or instructions for receiving a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

[0016] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

[0017] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

[0018] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a second capability of the one or more apparatuses to tune each RF chain of the plurality of RF chains to multiple subsets of the contiguous frequency bandwidth.

[0019] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the signal carries control information.

[0020] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, communicating the signal comprises receiving the signal; or transmitting the signal.

[0021] Another aspect provides a method for wireless communications by an apparatus. The method includes receiving, from a user equipment (UE), one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and communicating, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth.

[0022] Another aspect provides one or more apparatuses configured for wireless communications. The one or more apparatuses include one or more memories and one or more processors, coupled to the one or more memories, configured to cause the one or more apparatuses to receive, from a UE, one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and communicate, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth. For example, performance of any of the above steps may be by only one apparatus or by multiple apparatuses in a distributed fashion.

[0023] Another aspect provides one or more apparatuses configured for wireless communications. The one or more apparatuses include means for receiving, from a UE, one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and means for communicating, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth. For example, performance of any of the above steps may be by only one apparatus or by multiple apparatuses in a distributed fashion.

[0024] Another aspect provides one or more non-transitory computer-readable media. The one or more non-transitory computer-readable media include executable instructions that, when executed by one or more processors of one or more apparatuses, cause the one or more apparatuses to receive, from a UE, one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and communicate, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth. For example, the instructions may be included in only one computer-readable medium or in a distributed fashion across multiple computer-readable media, such that instructions may be executed by only one processor or by multiple processors in a distributed fashion, such that each apparatus of the one or more apparatuses may include one processor or multiple processors, and/or such that performance may be by only one apparatus or in a distributed fashion across multiple apparatuses.

[0025] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum FFT size per RF chain supported by the UE.

[0026] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum bandwidth per RF chain supported by the UE.

[0027] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the UE.

[0028] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a maximum number of MIMO layers supported by the UE when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

[0029] Some examples of the methods, apparatuses, and non-transitory computer-readable media described herein may further include operations, features, means, or instructions for determining a maximum number of MIMO layers supported by the UE when the UE is operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of: a maximum bandwidth; a SCS configured for communication between the one or more apparatuses and the UE; a number of RF chains of the UE configured for communication between the one or more apparatuses and the UE; a maximum FFT size per RF chain supported by the UE; a maximum bandwidth per RF chain supported by the UE; or a maximum number of subsets of contiguous frequency resources supported by the UE.

[0030] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the maximum bandwidth comprises a bandwidth configured for communication between the UE and the one or more apparatuses.

[0031] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the maximum bandwidth comprises a nominal maximum system bandwidth of the one or more apparatuses.

[0032] Some examples of the methods, apparatuses, and non-transitory computer-readable media described herein may further include operations, features, means, or instructions for transmitting a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

[0033] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

[0034] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

[0035] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the one or more reports further indicate a second capability of the UE to tune each RF chain of a plurality of RF chains of the UE to multiple subsets of the contiguous frequency bandwidth.

[0036] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, the signal carries control information.

[0037] In some examples of the methods, apparatuses, and non-transitory computer-readable media described herein, communicating the signal comprises: transmitting the signal; or receiving the signal.

[0038] An apparatus may comprise a processing system, a device with a processing system, or processing systems cooperating over one or more networks. An apparatus may comprise one or more memories; and one or more processors configured to cause the apparatus to perform any portion of any method described herein. In some examples, one or more of the processors may be preconfigured to perform various functions or operations described herein without requiring configuration by software.

[0039] The following description and the appended figures set forth certain features for purposes of illustration.

BRIEF DESCRIPTION OF DRAWINGS

[0040] The appended figures depict certain features of the various aspects described herein and are not to be considered limiting of the scope of this disclosure.

[0041] FIG. **1** depicts an example wireless communications network.

[0042] FIG. **2** depicts an example disaggregated base station architecture.

[0043] FIG. **3** depicts aspects of an example base station and an example user equipment (UE).

[0044] FIGS. **4**A, **4**B, **4**C, and **4**D depict various example aspects of data structures for a wireless communications network.

[0045] FIGS. **5**A and **5**B depict example processing at a UE using multiple radio frequency (RF) chains for communication.

[0046] FIG. **6** depicts a process flow for communications in a network between a network entity and a UE to report, at least, a multiple RF chains processing capability of the UE.

[0047] FIGS. **7**A and **7**B depict process flows for communications in a network between a network entity and a UE to determine a number of multiple-input-multiple-output (MIMO) layer(s) to use for downlink and uplink communications, respectively.

[0048] FIG. **8**A depicts a process flow for communications in a network between a network entity and a UE to perform reference signal (RS) measurements.

[0049] FIGS. **8**B and **8**C depict example RS configurations configuring one or more RS occasions.

[0050] FIG. **9** depicts a method for wireless communications.

[0051] FIG. **10** depicts another method for wireless communications.

[0052] FIG. **11** depicts aspects of an example communications device.

[0053] FIG. **12** depicts aspects of an example communications device.

DETAILED DESCRIPTION

[0054] Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer-readable mediums for reporting a multiple radio frequency (RF) chains processing capability of a UE to process, for example, wideband signals. As used herein, a wideband signal may be a signal communicated, via a carrier, over a relatively large bandwidth.

[0055] Every new generation of wireless technology brings new capabilities, such as increased carrier bandwidth to increase the total bandwidth available for data transmission. For example, the move from 3G to 4G grew carrier bandwidth size from 5 megahertz (MHz) to 20 MHz, while the transition from 4G to 5G saw carrier bandwidth grow from 20 MHz to 100 MHz. This increase in spectral bandwidth is similarly expected to continue for 6G, as well as other later-released generations of wireless technologies. For example, 6G is expected to expand the carrier bandwidth from 100 MHz (e.g., currently for 5G wireless technologies) to approximately 400 MHz to provide up to a four time increase in capacity. This level of capacity, which may be achieved in 6G, may help to extend the performance of 5G applications, and/or may help to expand the scope of capabilities needed to support new and innovative applications in wireless connectivity, cognition, sensing, and imaging. Further, the increased capacity in 6G may help to achieve peak data rates for data transfers.

[0056] While an increase in carrier bandwidth may provide the aforementioned advantages, in some cases, the increased carrier bandwidth may be greater than a single-carrier bandwidth capability of a UE. For example, in orthogonal frequency division multiplexing (OFDM) systems, Fast Fourier Transform (FFT), which is used to convert time domain signals to the frequency domain, and/or inverse FFT (iFFT), which is used to convert frequency domain signals to the time domain, may be implemented in a radio frequency (RF) chain of a UE, such as a receiver (RX) chain or transmitter (TX) chain of the UE, to receive or transmit an OFDM signal. In certain cases, the RF chain of the UE may be limited in a number of FFT or iFFT operations it can perform on a given communication, such as based on a design of the RF chain. For example, the RF chain may have processing constraints limiting how many FFT operations or iFFT operations it can perform within a time interval. The maximum number of FFT operations or iFFT operations that the RF chain can perform for a communication may be referred to as a maximum FFT size of the RF chain. For example, while a signal communicated in a time interval and across a carrier bandwidth of 100 MHz may require 4,096 FFT operations to be performed, a signal communicated in a time interval and across a carrier bandwidth of 400 MHz may require 16,384 FFT operations to be performed. Some UEs may not have RF chains with a maximum FFT size that is able to process such a signal communicated across a larger carrier bandwidth. As such, this FFT size limitation of some RF chains presents a technical problem for wireless technologies that support larger single carrier bandwidths.

[0057] In certain aspects, an RF chain that is a TX chain may convert digital baseband signals into analog RF signals. The RF chain that is a TX chain may include a baseband filter (BBF), a mixer (which may include one or several mixers), and a digital to analog converter (DAC). In certain aspects, an RF chain that is an RX chain may convert analog RF or intermediate frequency (IF) signals into baseband signals. The RF chain that is an RX chain may include a mixer (which may include one or several mixers), an analog to digital converter (ADC), and a BBF. A device with MIMO capability may have multiple RF chains for simultaneously processing multiple data streams and/or signals.

[0058] In certain aspects, to overcome this technical problem associated with increased carrier bandwidth, multiple RF chain processing at a UE may be enabled. For example, a UE may include multiple RF chains. Each RF chain may be coupled to one or more antenna elements or antenna arrays. RF chains may share antenna elements or arrays, or may be coupled to different antenna elements or arrays. Each RF chain may be configured to process signals received at the UE (e.g., from a network entity, another UE, etc.) and/or transmitted by the UE (e.g., to a network entity, another UE, etc.). In certain aspects, multiple RF chains at a UE may be used to process larger bandwidth signals (e.g., signals communicated over a contiguous bandwidth that are greater than a carrier bandwidth capability of a single RF chain of the UE based on the FFT size supported by the RF chain). In particular, each of multiple RF chains at the UE may be used to process a portion of a signal bandwidth (e.g., partial bandwidth/subband processing) to enable wideband operation. For example, each RF chain of the UE may support an FFT size capable of processing a 100 MHz bandwidth signal. To communicate across a larger contiguous frequency bandwidth, such as 400 MHz, the UE may be configured to use multiple RF chains, e.g., four RF chains, to communicate across the larger contiguous frequency bandwidth, where each RF chain is used communicate across a respective partial bandwidth of the larger contiguous frequency bandwidth. For example, the larger contiguous frequency bandwidth may be divided into a plurality of subsets of the contiguous frequency bandwidth, and each RF chain may be used to communicate across a respective subset of the contiguous frequency bandwidth. Accordingly, multiple RF chains processing at a UE may allow for the use of larger bandwidth carriers to increase capacity of the wireless communications system, without increasing a maximum FFT size per RF chain supported at a UE and/or without increasing signaling overhead (e.g., a disadvantage of using carrier aggregation, as described herein, such as based on carrier aggregation requiring guard bands between carriers, such that the carriers are not contiguous in frequency).

[0059] In certain aspects, increased capacity achieved via multiple RF chains processing at a UE may be at the expense of a multiple-input-multiple-output (MIMO) operation implemented between the UE and a wireless communications device (e.g., a network entity or another UE). MIMO is a wireless technology that uses multiple antennas at both a transmitter and a receiver to transmit and receive multiple independent data streams, also referred to as "layers," simultaneously to improve communication performance by exploiting spatial diversity and multipath propagation. Specifically, MIMO layers refer to the parallel spatial channels in a MIMO system, which play a crucial role in spatial multiplexing, leading to increased data rates and improved system performance. In certain aspects, a UE's MIMO capability, or a number of MIMO layers that may be used for MIMO communication (e.g., uplink and/or downlink communication) at the UE, may be reduced based on a number of RF chains being used to process signal(s) across a given carrier bandwidth at the UE. For example, each MIMO layer may require separate one or more RF chains to process the MIMO layer, where the one or more RF chains required to process a given MIMO layer is based on the carrier bandwidth (C_B) used for communication, and a maximum bandwidth (B_{MAX}) the UE can process per RF chain. For example, to process L MIMO layers, the UE may require R RF chains, where $R=L*C_B/B_{MAX}$. Accordingly, if R and B_{MAX} remain constant, as C_B increases, L decreases.

[0060] Due to the impact of multiple RF chains processing on MIMO operations, and more specifically on a number of MIMO layers that may be used for communication, wireless communications devices, such as a network entity, may need to be informed about a UE's capability to use multiple RF chains at a time when communicating with the UE. Thus, aspects described herein provide techniques for reporting a UE's multiple RF chains processing capability to wireless communications devices, such as a network entity. For example, a UE may send, to a network entity, report(s) indicating a capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources, and more specifically, a capability of the UE to use each of the multiple RF chains to communicate across a subset of the contiguous frequency resources (e.g., a subband of a bandwidth used for communication). The indication may be a simple binary number indicating whether or not the UE is capable of using multiple RF chains at a time to communicate across contiguous frequency resources, or some other indication. The network entity receiving the report(s), comprising such capability information, may use this information when communicating with the UE. For example, the network entity may use this information when determining a number of MIMO layers to use for downlink or uplink communications with the UE.

[0061] In certain aspects, in addition to reporting a multiple RF chains processing capability of the UE, the UE may report additional capability information to a wireless communications device, such as a network entity to aid the network entity in determining a number of MIMO layers to use for communications with the UE. This capability information may include information about a maximum FFT size per RF chain supported by the UE, information about a maximum number of subsets of contiguous frequency resources (e.g., subbands) supported by UE, and/or the like. Alternatively, the capability information may include an explicit indication of a maximum number of MIMO layers supported at the UE when using multiple RF chains to communicate across contiguous frequency resources.

[0062] In certain aspects, different RF antenna elements (or arrays), associated with RF chains at the UE, may not be co-located at the UE. Thus, some RF chains may be better suited for different frequency subbands than other RF chains at the UE. To determine which subband(s) to tune each RF chain at the UE to, one or more reference signals (RSs) may be measured by one or more RF chains at the UE for a contiguous frequency bandwidth and/or for one or subbands of the contiguous frequency bandwidth. To enable such measurement, different configurations of RS occasion(s) configured for communicating RS(s) spanning the contiguous frequency bandwidth, may be configured at the UE.

Introduction to Wireless Communications Networks

[0063] The techniques and methods described herein may be used for various wireless communications networks. While aspects may be described herein using terminology commonly associated with 3G, 4G, 5G, 6G, and/or other generations of wireless technologies, aspects of the present disclosure may likewise be applicable to other communications systems and standards not explicitly mentioned herein.

[0064] FIG. **1** depicts an example of a wireless communications network **100**, in which aspects described herein may be implemented.

[0065] Generally, wireless communications network **100** includes various network entities (alternatively, network elements or network nodes). A network entity is generally a communications device and/or a communications function performed by a communications device (e.g., a user equipment (UE), a base station (BS), a component of a BS, a server, etc.). As such communications devices are part of wireless communications network **100**, and facilitate wireless communications, such communications devices may be referred to as wireless communications devices. For example, various functions of a network as well as various devices associated with and interacting with a network may be considered network entities. Further, wireless communications network **100** includes terrestrial aspects, such as ground-based network entities (e.g., BSs **102**), and non-terrestrial aspects (also referred to herein as non-terrestrial network entities), such as satellite **140** and/or aerial or spaceborne platform(s), which may include network entities on-board (e.g., one or more BSs) capable of communicating with other network elements (e.g., terrestrial BSs) and UEs.

[0066] In the depicted example, wireless communications network **100** includes BSs **102**, UEs **104**, and one or more core networks, such as an Evolved Packet Core (EPC) **160** and 5G Core (5GC) network **190**, which interoperate to provide communications services over various communications links, including wired and wireless links.

[0067] FIG. **1** depicts various example UEs **104**, which may more generally include: a cellular phone, smart phone, session initiation protocol (SIP) phone, laptop, personal digital assistant (PDA), satellite radio, global positioning system, multimedia device, video device, digital audio player, camera, game console, tablet, smart device, wearable device, vehicle, electric meter, gas pump, large or small kitchen appliance, healthcare device, implant, sensor/actuator, display, internet of things (IoT) devices, always on (AON) devices, edge processing devices, data centers, or other similar devices. UEs **104** may also be referred to more generally as a mobile device, a wireless device, a station, a mobile station, a subscriber station, a mobile subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a remote device, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, and others.

[0068] BSs **102** wirelessly communicate with (e.g., transmit signals to or receive signals from) UEs **104** via communications links **120**. The communications links **120** between BSs **102** and UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to a BS **102** and/or downlink (DL) (also referred to as forward link) transmissions from a BS **102** to a UE **104**. The communications links **120** may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity in various aspects.

[0069] BSs **102** may generally include: a NodeB, enhanced NodeB (eNB), next generation enhanced NodeB (ng-eNB), next generation NodeB (gNB or gNodeB), access point, base transceiver station, radio base station, radio transceiver, transceiver function, transmission reception point, and/or others. Each of BSs **102** may provide communications coverage for a respective coverage area **110**, which may sometimes be referred to as a cell, and which may overlap in some cases (e.g., small cell **102'** may have a coverage area **110'** that overlaps the coverage area **110** of a macro cell). A BS may, for example, provide communications coverage for a macro cell (covering relatively large geographic area), a pico cell (covering relatively smaller geographic area, such as a sports stadium), a femto cell (relatively smaller geographic area (e.g., a home)), and/or other types of cells.

[0070] Generally, a cell may refer to a portion, partition, or segment of wireless communication coverage served by a network entity within a wireless communication network. A cell may have geographic characteristics, such as a geographic coverage area, as well as radio frequency characteristics, such as time and/or frequency resources dedicated to the cell. For example, a specific geographic coverage area may be covered by multiple cells employing different frequency resources (e.g., bandwidth parts) and/or different time resources. As another example, a specific geographic coverage area may be covered by a single cell. In some contexts (e.g., a carrier aggregation scenario and/or multi-connectivity scenario), the terms "cell" or "serving cell" may refer to or correspond to a specific carrier frequency (e.g., a component carrier) used for wireless communications, and a "cell group" may refer to or correspond to multiple carriers used for wireless communications. As examples, in a carrier aggregation scenario, a UE may communicate on multiple component carriers corresponding to multiple (serving) cells in the same cell group, and in a multi-connectivity (e.g., dual connectivity) scenario, a UE may communicate on multiple component carriers corresponding to multiple cell groups.

[0071] While BSs **102** are depicted in various aspects as unitary communications devices, BSs **102** may be implemented in various configurations. For example, one or more components of a base station may be disaggregated, including a central unit (CU), one or more distributed units (DUs), one or more radio units (RUs), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, to name a few examples. In another example, various aspects of a base station may be virtualized. More generally, a base station (e.g., BS **102**) may include components that are located at a single physical location or components located at various physical locations. In examples in which a base station includes components that are located at various physical locations, the various components may each perform functions such that, collectively, the various components achieve functionality that is similar to a base station that is located at a single physical location. In some aspects, a base station including components that are located at various physical locations may be referred to as a disaggregated radio access network architecture, such as an Open RAN (O-RAN) or Virtualized RAN (VRAN) architecture. FIG. **2** depicts and describes an example disaggregated base station architecture.

[0072] Different BSs **102** within wireless communications network **100** may also be configured to support different radio access technologies, such as 3G, 4G, and/or 5G. For example, BSs **102** configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC **160** through first backhaul links **132** (e.g., an S1 interface). BSs **102** configured for 5G (e.g., 5G NR or Next Generation RAN (NG-RAN)) may interface with 5GC **190** through second backhaul links **184**. BSs **102** may communicate directly or indirectly (e.g., through the EPC **160** or 5GC **190**) with each other over third backhaul links **134** (e.g., X2 interface), which may be wired or wireless.

[0073] Wireless communications network **100** may subdivide the electromagnetic spectrum into various classes, bands, channels, or other features. In some aspects, the subdivision is provided based on wavelength and frequency, where frequency may also be referred to as a carrier, a subcarrier, a frequency channel, a tone, or a subband. For example, 3GPP currently defines Frequency Range 1 (FR1) as including 410 MHz-7125 MHz, which is often referred to (interchangeably) as "Sub-6 GHz". Similarly, 3GPP currently defines Frequency Range 2 (FR2) as including 24,250 MHz-71,000 MHz, which is sometimes referred to (interchangeably) as a "millimeter wave" ("mmW" or "mmWave"). In some cases, FR2 may be further defined in terms of sub-ranges, such as a first sub-range FR2-1 including 24,250 MHz-52,600 MHz and a second sub-range FR2-2 including 52,600 MHz-71,000 MHz. A base station configured to communicate using mm Wave/near mm Wave radio frequency bands (e.g., a mmWave base station such as BS **180**) may utilize beamforming (e.g., **182**) with a UE (e.g., **104**) to improve path loss and range.

[0074] The communications links **120** between BSs **102** and, for example, UEs **104**, may be through one or more carriers, which may have different bandwidths (e.g., 5, 10, 15, 20, 100, 400, and/or other MHz), and which may be aggregated in various aspects. Carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL).

[0075] Communications using higher frequency bands may have higher path loss and a shorter range compared to lower frequency communications. Accordingly, certain base stations (e.g., **180** in FIG. **1**) may utilize beamforming **182** with a UE **104** to improve path loss and range. For example, BS **180** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate the beamforming. In some cases, BS **180** may transmit a beamformed signal to UE **104** in one or more transmit directions **182'**. UE **104** may receive the beamformed signal from the BS **180** in one or more receive directions **182"**. UE **104** may also transmit a beamformed signal to the BS **180** in one or more transmit directions **182"**. BS **180** may also receive the beamformed signal from UE **104** in one or more receive directions **182'**. BS **180** and UE **104** may then perform beam training to determine the best receive and transmit directions for each of BS **180** and UE **104**. Notably, the transmit and receive directions for BS **180** may or may not be the same. Similarly, the transmit and receive directions for UE **104** may or may not be the same.

[0076] Wireless communications network **100** further includes a Wi-Fi AP **150** in communication with Wi-Fi stations (STAs) **152** via communications links **154** in, for example, a 2.4 GHz and/or 5 GHz unlicensed frequency spectrum.

[0077] Certain UEs **104** may communicate with each other using device-to-device (D2D) communications link **158**. D2D communications link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel

(PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), a physical sidelink control channel (PSCCH), and/or a physical sidelink feedback channel (PSFCH).

[0078] EPC **160** may include various functional components, including: a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and/or a Packet Data Network (PDN) Gateway **172**, such as in the depicted example. MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, MME **162** provides bearer and connection management.

[0079] Generally, user Internet protocol (IP) packets are transferred through Serving Gateway **166**, which itself is connected to PDN Gateway **172**. PDN Gateway **172** provides UE IP address allocation as well as other functions. PDN Gateway **172** and the BM-SC **170** are connected to IP Services **176**, which may include, for example, the Internet, an intranet, an IP Multimedia Subsystem (IMS), a Packet Switched (PS) streaming service, and/or other IP services.

[0080] BM-SC **170** may provide functions for MBMS user service provisioning and delivery. BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and/or may be used to schedule MBMS transmissions. MBMS Gateway **168** may be used to distribute MBMS traffic to the BSs **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and/or may be responsible for session management (start/stop) and for collecting eMBMS related charging information.

[0081] 5GC **190** may include various functional components, including: an Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a Session Management Function (SMF) **194**, and a User Plane Function (UPF) **195**. AMF **192** may be in communication with Unified Data Management (UDM) **196**.

[0082] AMF **192** is a control node that processes signaling between UEs **104** and 5GC **190**. AMF **192** provides, for example, quality of service (QoS) flow and session management.

[0083] Internet protocol (IP) packets are transferred through UPF **195**, which is connected to the IP Services **197**, and which provides UE IP address allocation as well as other functions for 5GC **190**. IP Services **197** may include, for example, the Internet, an intranet, an IMS, a PS streaming service, and/or other IP services.

[0084] In various aspects, a network entity or network node can be implemented as an aggregated base station, as a disaggregated base station, a component of a base station, an integrated access and backhaul (IAB) node, a relay node, a sidelink node, to name a few examples.

[0085] FIG. **2** depicts an example disaggregated base station **200** architecture. The disaggregated base station **200** architecture may include one or more central units (CUs) **210** that can communicate directly with a core network **220** via a backhaul link, or indirectly with the core network **220** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **225** via an E2 link, or a Non-Real Time (Non-RT) RIC **215** associated with a Service Management and Orchestration (SMO) Framework **205**, or both). A CU **210** may communicate with one or more distributed units (DUs) **230** via respective midhaul links, such as an F1 interface. The DUs **230** may communicate with one or more radio units (RUs) **240** via respective fronthaul links. The RUs **240** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **240**.

[0086] Each of the units, e.g., the CUS **210**, the DUs **230**, the RUs **240**, as well as the Near-RT RICs **225**, the Non-RT RICs **215** and the SMO Framework **205**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communications interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally or alternatively, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a radio frequency (RF) transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0087] In some aspects, the CU **210** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **210**. The CU **210** may be configured to handle user plane functionality (e.g., Central Unit-User Plane (CU-UP)), control plane functionality (e.g., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **210** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the El interface when implemented in an O-RAN configuration. The CU **210** can be implemented to communicate with the DU **230**, as necessary, for network control and signaling.

[0088] The DU **230** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **240**. In some aspects, the DU **230** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3*rd* Generation Partnership Project (3GPP). In some aspects, the DU **230** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **230**, or with the control functions hosted by the CU **210**.

[0089] Lower-layer functionality can be implemented by one or more RUs **240**. In some deployments, an RU **240**, controlled by a DU **230**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **240** can be implemented to handle over the air (OTA) communications with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communications with the RU(s) **240** can be controlled by the corresponding DU **230**. In some scenarios, this configuration can enable the DU(s) **230** and the CU **210** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0090] The SMO Framework **205** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **205** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **205** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **290**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **210**, DUs **230**, RUs **240** and Near-RT RICs **225**. In some implementations, the SMO Framework **205** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **211**, via an O1 interface. Additionally, in some implementations, the SMO Framework **205** can communicate directly with one or more DUs **230** and/or one or more RUs **240** via an O1 interface. The SMO Framework **205** also may include a Non-RT RIC **215** configured to support functionality of the SMO Framework **205**.

[0091] The Non-RT RIC **215** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **225**. The Non-RT RIC **215** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **225**. The Near-RT RIC **225** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **210**, one or more DUs **230**, or both, as well as an O-eNB, with the Near-RT RIC **225**.

[0092] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **225**, the Non-RT RIC **215** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **225** and may be received at the SMO Framework **205** or the Non-RT RIC **215** from non-network data sources or from network functions. In some examples, the Non-RT RIC **215** or the Near-RT RIC **225** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **215** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **205** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

[0093] FIG. **3** depicts aspects of an example BS **102** and a UE **104**.

[0094] Generally, BS **102** includes various processors (e.g., **318**, **320**, **330**, **338**, and **340**), antennas **334***a-t* (collectively **334**), transceivers **332***a-t* (collectively **332**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., data source **312**) and wireless reception of data (e.g., data sink **314**). For example, BS **102** may send and receive data between BS **102** and UE **104**. BS **102** includes controller/processor **340**, which may be configured to implement various functions described herein related to wireless communications. Note that the BS **102** may have a disaggregated architecture as described herein with respect to FIG. **2**.

[0095] Generally, UE **104** includes various processors (e.g., **358**, **364**, **366**, **370**, and **380**), antennas **352***a-r* (collectively **352**), transceivers **354***a-r* (collectively **354**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., retrieved from data source **362**) and wireless reception of data (e.g., provided to data sink **360**). UE **104** includes controller/processor **380**, which may be configured to implement various functions described herein related to wireless communications.

[0096] In regards to an example downlink transmission, BS **102** includes a transmit processor **320** that may receive data from a data source **312** and control information from a controller/processor **340**. The control information may be for the physical broadcast channel (PBCH), physical control format indicator channel (PCFICH), physical hybrid automatic repeat request (HARQ) indicator channel (PHICH), physical downlink control channel (PDCCH), group common PDCCH (GC PDCCH), and/or others. The data may be for the physical downlink shared channel (PDSCH), in some examples.

[0097] Transmit processor **320** may process (e.g., encode and symbol map) the data and control information to obtain data symbols and control symbols, respectively. Transmit processor **320** may also generate reference symbols, such as for the primary synchronization signal (PSS), secondary synchronization signal (SSS), PBCH demodulation reference signal (DMRS), and channel state information reference signal (CSI-RS).

[0098] Transmit (TX) multiple-input multiple-output (MIMO) processor **330** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, and/or the reference symbols, if applicable, and may provide output symbol streams to the modulators (MODs) in transceivers **332***a*-**332***t*. Each modulator in transceivers **332***a*-**332***t* may process a respective output symbol stream to obtain an output sample stream. Each modulator may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a downlink signal. Downlink signals from the modulators in transceivers **332***a*-**332***t* may be transmitted via the antennas **334***a*-**334***t*, respectively.

[0099] In order to receive the downlink transmission, UE **104** includes antennas **352***a*-**352***r* that may receive the downlink signals from the BS **102** and may provide received signals to the demodulators (DEMODs) in transceivers **354***a*-**354***r*, respectively. Each demodulator in transceivers **354***a*-**354***r* may condition (e.g., filter, amplify, downconvert, and digitize) a respective received signal to obtain input samples. Each demodulator may further process the input samples to obtain received symbols.

[0100] RX MIMO detector **356** may obtain received symbols from all the demodulators in transceivers **354***a*-**354***r*, perform MIMO detection on the received symbols if applicable, and provide detected symbols. Receive processor **358** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, provide decoded data for the UE **104** to a data sink **360**, and provide decoded control information to a controller/processor **380**.

[0101] In regards to an example uplink transmission, UE **104** further includes a transmit processor **364** that may receive and process data (e.g., for the PUSCH) from a data source **362** and control information (e.g., for the physical uplink control channel (PUCCH)) from the controller/processor **380**. Transmit processor **364** may also generate reference symbols for a reference signal (e.g., for the sounding reference signal (SRS)). The symbols from the transmit processor **364** may be precoded by a TX MIMO processor **366** if applicable, further processed by the modulators in transceivers **354***a*-**354***r* (e.g., for SC-FDM), and transmitted to BS **102**.

[0102] At BS **102**, the uplink signals from UE **104** may be received by antennas **334***a-t*, processed by the demodulators in transceivers **332***a*-**332***t*, detected by a RX MIMO detector **336** if applicable, and further processed by a receive processor **338** to obtain decoded data and control information sent by UE **104**. Receive processor **338** may provide the decoded data to a data sink **314** and the decoded control information to the controller/processor **340**.

[0103] Memories **342** and **382** may store data and program codes for BS **102** and UE **104**, respectively.

[0104] Scheduler **344** may schedule UEs for data transmission on the downlink and/or uplink.

[0105] In various aspects, BS **102** may be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, "transmitting" may refer to various mechanisms of outputting data, such as outputting data from data source **312**, scheduler **344**, memory **342**, transmit processor **320**, controller/processor **340**, TX MIMO processor **330**, transceivers **332***a-t*, antenna **334***a-t*, and/or other aspects described herein. Similarly, "receiving" may refer to various mechanisms of obtaining data, such as obtaining data from antennas **334***a-t*, transceivers **332***a-t*, RX MIMO detector **336**, controller/processor **340**, receive processor **338**, scheduler **344**, memory **342**, and/or other aspects described herein.

[0106] In various aspects, UE **104** may likewise be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, "transmitting" may refer to various mechanisms of outputting data, such as outputting data from data source **362**, memory **382**, transmit processor **364**, controller/processor **380**, TX MIMO processor **366**, transceivers **354***a-t*, antenna **352***a-t*, and/or other aspects described herein. Similarly, "receiving" may refer to various mechanisms of obtaining data, such as obtaining data from antennas **352***a-t*, transceivers **354***a-t*, RX MIMO detector **356**, controller/processor **380**, receive processor **358**, memory **382**, and/or other aspects described herein.

[0107] In some aspects, a processor may be configured to perform various operations, such as those associated with the methods described herein, and transmit (output) to or receive (obtain) data from another interface that is configured to transmit or receive, respectively, the data.

[0108] In various aspects, artificial intelligence (AI) processors **318** and **370** may perform AI processing for BS **102** and/or UE **104**, respectively. The AI processor **318** may include AI accelerator hardware or circuitry such as one or more neural processing units (NPUs), one or more neural network processors, one or more tensor processors, one or more deep learning processors, etc. The AI processor **370** may likewise include AI accelerator hardware or circuitry. As an example, the AI processor **370** may perform AI-based beam management, AI-based channel state feedback (CSF), AI-based antenna tuning, and/or AI-based positioning (e.g., non-line of sight positioning prediction). In some cases, the AI processor **318** may process feedback from the UE **104** (e.g., CSF) using hardware accelerated AI inferences and/or AI training. The AI processor **318** may decode compressed CSF from the UE **104**, for example, using a hardware accelerated AI inference associated with the CSF. In certain cases, the AI processor **318** may perform certain RAN-based functions including, for example, network planning, network performance management, energy-efficient network operations, etc.

[0109] FIGS. **4**A, **4**B, **4**C, and **4**D depict aspects of data structures for a wireless communications network, such as wireless communications network **100** of FIG. **1**.

[0110] In particular, FIG. **4**A is a diagram **400** illustrating an example of a first subframe within a 5G (e.g., 5G NR) frame structure, FIG. **4**B is a diagram **430** illustrating an example of DL channels within a 5G subframe, FIG. **4**C is a diagram **450** illustrating an example of a second subframe within a 5G frame structure, and FIG. **4**D is a diagram **480** illustrating an example of UL channels within a 5G subframe.

[0111] Wireless communications systems may utilize orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) on the uplink and downlink. Such systems may also support half-duplex operation using time division duplexing (TDD). OFDM and single-carrier frequency division multiplexing (SC-FDM) partition the system bandwidth (e.g., as depicted in FIGS. **4**B and **4**D) into multiple orthogonal subcarriers. Each subcarrier may be modulated with data. Modulation symbols may be sent in the frequency domain with OFDM and/or in the time domain with SC-FDM.

[0112] A wireless communications frame structure may be frequency division duplex (FDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for either DL or UL. Wireless communications frame structures may also be time division duplex (TDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for both DL and UL.

[0113] In FIG. **4**A and **4**C, the wireless communications frame structure is TDD where D is DL, U is UL, and X is flexible for use between DL/UL. UEs may be configured with a slot format through a received slot format indicator (SFI) (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling). In the depicted examples, a 10 ms frame is divided into 10 equally sized 1 ms subframes. Each subframe may include one or more time slots. In some examples, each slot may include 12 or 14 symbols, depending on the cyclic prefix (CP) type (e.g., 12 symbols per slot for an extended CP or 14 symbols per slot for a normal CP).

Subframes may also include mini-slots, which generally have fewer symbols than an entire slot. Other wireless communications technologies may have a different frame structure and/or different channels.

[0114] In certain aspects, the number of slots within a subframe (e.g., a slot duration in a subframe) is based on a numerology, which may define a frequency domain subcarrier spacing and symbol duration as further described herein. In certain aspects, given a numerology μ, there are 2^{μ} slots per subframe. Thus, numerologies (μ) 0 to 6 may allow for 1, 2, 4, 8, 16, 32, and 64 slots, respectively, per subframe. In some cases, the extended CP (e.g., 12 symbols per slot) may be used with a specific numerology, e.g., numerology 2 allowing for 4 slots per subframe. The subcarrier spacing and symbol length/duration are a function of the numerology. The subcarrier spacing may be equal to $2^{\mu}\times15$ kHz, where u is the numerology **0** to 6. As an example, the numerology μ=0 corresponds to a subcarrier spacing of 15 kHz, and the numerology μ=6 corresponds to a subcarrier spacing of 960 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **4**A, **4**B, **4**C, and **4**D provide an example of a slot format having 14 symbols per slot (e.g., a normal CP) and a numerology μ=2 with 4 slots per subframe. In such a case, the slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 µs.

[0115] As depicted in FIGS. **4**A, **4**B, **4**C, and **4**D, a resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends, for example, 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme including, for example, quadrature phase shift keying (QPSK) or quadrature amplitude modulation (QAM).

[0116] As illustrated in FIG. **4**A, some of the REs carry reference (pilot) signals (RS) for a UE (e.g., UE **104** of FIGS. **1** and **3**). The RS may include demodulation RS (DMRS) and/or channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and/or phase tracking RS (PT-RS).

[0117] FIG. **4**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs), each CCE including, for example, nine RE groups (REGs), each REG including, for example, four consecutive REs in an OFDM symbol.

[0118] A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE (e.g., **104** of FIGS. **1** and **3**) to determine subframe/symbol timing and a physical layer identity.

[0119] A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing.

[0120] Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the aforementioned DMRS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (SSB), and in some cases, referred to as a synchronization signal block (SSB). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and/or paging messages.

[0121] As illustrated in FIG. **4**C, some of the REs carry DMRS (indicated as R for one particular configuration, but other DMRS configurations are possible) for channel estimation at the base station. The UE may transmit DMRS for the PUCCH and DMRS for the PUSCH. The PUSCH DMRS may be transmitted, for example, in the first one or two symbols of the PUSCH. The PUCCH DMRS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. UE **104** may transmit sounding reference signals (SRS). The SRS may be transmitted, for example, in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

[0122] FIG. **4**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and HARQ ACK/NACK feedback. The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

Aspects Related to Carrier Bandwidth

[0123] As described herein, network entities (e.g., such as BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**) wirelessly communicate with (e.g., transmit signals to or receive signals from) UEs (e.g., such as UE **104** depicted and described with respect to FIGS. **1** and **3**) via communications links (e.g., such as communication links **120** depicted and described with respect to FIG. **1**). The communications may occur using one or more carriers. For example, a network entity may use one or more carriers associated with a cell of the network entity for communications with UE(s) in the cell. Similarly, a UE may use one or more carriers associated with the cell for communications with the network entity in the cell.

[0124] Carriers used for such communication may have different bandwidths. Bandwidth, in spectrum, refers to a range of frequencies used to convey signals that transmit data over the radio frequency. The more bandwidth assigned to a carrier, the more data it can send and receive at one time. A bandwidth of an individual carrier may be equal to 5, 10, 15, 20, 100, and/or other MHz, up to a maximum carrier bandwidth defined for various wireless technologies.

[0125] For example, different maximum carrier bandwidths are defined for 2G, 3G, 4G, and 5G wireless technologies. For 2G, a maximum bandwidth of an individual 2G carrier is 200 kHz (e.g., per Global System for Mobile Communications (GSM)) or 1.25 MHz (e.g., per Interim Standards 95 (IS-95)). For 3G, a maximum bandwidth of an individual 3G carrier is 5 MHz, or four times the carrier size of a 2G carrier. For 4G and 5G, a maximum bandwidth of an individual carrier is 20 MHz (e.g., four times greater than the carrier size of a 3G carrier) and 100 MHz (e.g., five times greater than the carrier size of a 4G carrier), respectively. Thus, in essence, every new generation of wireless technology has increased carrier bandwidth from the previous generation. The increase in bandwidth increases the volume of data that can be communicated between entities at a time, thereby increasing the overall capacity of a wireless communications network. For example, a higher bandwidth implies that a larger amount of data can be transmitted, thus increasing the capacity of a cell.

[0126] This increase in spectral bandwidth is expected to continue for 6G, as well as other later-released generations of wireless technologies. Specifically, 6G is expected to expand the carrier bandwidth from 100 MHz (e.g., currently for 5G wireless technologies) to approximately 400 MHz (or 500 MHz) to provide up to a four times increase (or five times increase) in capacity. Further, in 6G, improvements in antennas, through the application of more sophisticated MIMO techniques may also help to make a significant increase in capacity for 6G wireless technologies by sending more streams of data simultaneously. This level of capacity offered in 6G may help to extend the performance of 5G applications, as well as expand the scope of capabilities to support new and innovative applications in wireless connectivity, cognition, sensing, and imaging. Further, the increased capacity in 6G may help to achieve peak data rates for data transfers.

[0127] Increasing carrier bandwidth in 6G and later generations of wireless technologies may increase the number of tones in a single carrier. For example, in 5G, the maximum carrier bandwidth is 100 MHz and is defined as 273 PRBs for a subcarrier spacing (SCS) of 30 kHz. As described above, each PRB may extend 12 consecutive subcarriers; thus, the 100 MHz bandwidth may include 3,276 subcarriers (e.g., 273 PRBs×12subcarriers per PRB=3,276 subcarriers), which may also be referred to as tones. Thus, a single 100 MHz carrier may include 3,276 tones.

[0128] In 6G, this number of tones per carrier may be increased by a factor of four. Specifically, as described herein, a maximum carrier bandwidth considered for 6G wireless technologies is 400 MHz. Thus, because a single 100 MHz carrier may include 3,276 tones, a single 400 MHz carrier, supported in 6G, may include a total of 13,104tones (e.g., 3,276 tones×4 scaling factor=13,104 tones).

[0129] In OFDM systems, an OFDM signal is generated by implementing, at a transmitter RF chain of a device (such as a UE or network entity), Inverse Fast Fourier Transform (IFFT), which is used to convert frequency domain to time domain. An OFDM signal may be decoded, at a receiver RF chain of a device, by implementing Fast Fourier Transform (FFT), which is used to convert time domain to frequency domain. As discussed, an FFT size of an RF chain of the UE may be limited, thereby limiting a maximum number of subcarriers, or tones, which may be processed by the RF chain. For example, 5G NR supports a maximum FFT size of 4K (4,096 FFT size). As such, a maximum number of subcarriers, or tones, which may be processed by a receiver of a signal, may be equal to 4,096 subcarriers. This FFT size limitation presents a technical problem for processing a larger number of subcarriers, such as for 6G wireless technologies, which may support a single 400 MHz carrier, including 13,104 tones. For example, in conventional transmitter and receiver architecture, an FFT size greater than 13K, such as 16K FFT, may be needed to process the 13,104 tones included in the 400 MHz carrier. Unfortunately, an FFT size greater than 4K may not be supported by some RF chains of some UEs.

[0130] One solution to this technical problem is to implement carrier aggregation. Carrier aggregation is a bandwidth-extension technology supported since LTE-Advanced, which can aggregate multiple carriers for simultaneous reception or transmission. Carrier aggregation may involve using multiple carriers simultaneously to create a wider channel for data transmission, which may help to increase data throughput and/or reduce latency, thereby allowing for a more efficient and responsive mobile network. For example, if a network entity supports the use of a 100 MHz carrier while a UE only supports up to a 25 MHz carrier (e.g., a "wide carrier" operation where carrier bandwidth is greater than single-carrier bandwidth capability for the UE), the UE may use intra-band carrier aggregation to implement a wideband operation of 100 MHz. Intra-band carrier aggregation involves aggregating carriers in a same frequency band, separated by a guard band (e.g., a frequency band used to prevent interference), for communication across a larger bandwidth. Increased capacity and higher data rates offered by carrier aggregation, however, comes at the expense of increased signaling overhead. For example, guard band overhead due to the placement of guard bands between each pair of carriers, more control signaling overhead, and/or the like associated with intra-band carrier aggregation, may contribute to the increased signaling overhead. In particular, in carrier aggregation, signals cannot be communicated in the guard bands, in order for the signals in the separate carriers to be decodable, and therefore, the communication is not over the entire contiguous bandwidth, but rather on separated bandwidths. For example, with carrier aggregation, to communicate over 100 MHz, the 100 MHz may be divided into separate carriers that are separated by guard bands. For example, there may be three guard bands of 4 MHz separating four carriers of 22 MHz, over the 100 MHz bandwidth, such that only 88 MHz is used for communication, instead of the entire contiguous 100 MHz.

[0131] Further, use of carrier aggregation for communication of a wideband signal contradicts the increase in carrier bandwidth, which may be supported by some wireless technologies. Thus, carrier aggregation operation for applications requiring wideband (e.g., larger) bandwidth operation, such as sensing, may not be desirable.

Aspects Related to Wideband Operation Using Multiple RF Chains

[0132] In certain aspects, to overcome the aforementioned technical problems associated with increased carrier bandwidth, which may be supported in 6G and later wireless generation technologies, multiple RF chain processing at a UE may be enabled. As described herein, an RF chain is a series of interconnected components, designed to process, receive, and/or transmit signals. Each RF chain may be configured to perform low-pass filtering (LPF), digital to analog conversion (DAC) or analog to digital conversion (ADC), and FFT or IFFT, to process signals.

[0133] In some cases, multiple RF chains at a UE may be used by the UE to process larger bandwidth signals (e.g., signals communicated over a bandwidth that is greater than a single RF chain of the UE can process). In particular, each RF chain at the UE may be used to process a portion of a signal bandwidth (e.g., partial bandwidth processing) to enable wideband operation in the uplink and/or the downlink. For example, a single RF chain of the UE may support processing of a 100 MHz carrier, such as for uplink or downlink communications with a network entity. Thus, for the UE to communicate across a contiguous frequency 400 MHz bandwidth using a 400 MHz carrier, the UE may use multiple RF chains. Specifically, four RF chains may be used, where each RF chain processes a portion of the signal bandwidth (e.g., a subband or a subset of the contiguous frequency bandwidth), for example 100 MHz of the 400 MHz bandwidth signal, to enable wideband operation.

[0134] Multiple RF chains processing at the UE may be enabled for uplink and/or downlink communications. In certain aspects, multiple RF chains processing is enabled at the UE for downlink communications, but not uplink communications, due to more demanding bandwidth in the downlink.

[0135] Further, in certain aspects, multiple RF chains processing at the UE may be enabled for only control information transmissions (e.g., non-data transfer signals), such as used for positioning, sensing, and/or the like. For example, in certain aspects, for data communication via the PDSCH and/or PUSCH, wideband operation at the UE may not be supported. Alternatively, in certain aspects, multiple RF chains processing at the UE may be enabled for all communication types, including both control signaling and data transmissions. For example, multiple RF chains processing at the UE may be used for data transfer-related signals/channels, which may help to improve communication peak performance (e.g., for downlink).

[0136] FIGS. **5**A and **5**B depict example processing **500***a,* **500***b* at a UE using multiple RF chains for communication. More specifically, example processing **500***a* depicted in FIG. **5**A utilizes multiple RF chains at the UE to receive a wideband signal from a network entity. Example processing **500***b* depicted in FIG. **5**B utilizes multiple RF chains at the UE to transmit a wideband signal to a network entity.

[0137] As shown in FIG. **5**A, a network entity **502** (e.g., an example of the BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**) sends, to a UE **504** (e.g., an example of the UE **104** depicted and described with respect to FIGS. **1** and **3**), a downlink signal **506** across a contiguous frequency bandwidth. Downlink signal **506** may be an RF signal or an intermediate frequency (IF) signal. For this example, network entity **502** may send downlink signal **506** to UE **504** across a 400 MHz bandwidth of contiguous frequency resources. However, in other examples, the downlink signal **506** may be sent across a different bandwidth.

[0138] UE **504** may support a maximum carrier bandwidth, such as of 100 MHz in an example, per RF chain **510**; thus, in this example, UE **504** may simultaneously use four RF chains **510-1**, **510-2**, **510-3**, and **510-4** (collectively referred to herein as "RF chains **510**" and individually referred to herein as "RF chain **510**") at UE **504** to receive and process the 400 MHz signal **506** from network entity **502** (e.g., 400 MHz bandwidth signal/100 MHz=4 RF chains). Specifically, each RF chain **510** may be used to process a subset of contiguous frequency resources, e.g., a 100 MHz subband, of the 400 MHz bandwidth signal **506**. The subbands may be non-overlapping in frequency and cover the entire bandwidth of signal **506**. Each RF chain **510** processes 100 MHz of the 400 MHz bandwidth signal **506** by (1) applying a 100 MHz LPF towards the 100 MHz bandwidth processed by the respective RF chain **510**, (2) performing ADC, and/or (3) performing FFT. LPF may be used to pass signals below a cutoff frequency while attenuating all signals above. As such, in this case, LPF performed at a single RF chain **510** may be used to obtain a narrow band (NB) signal of 100 MHz from the transmitted 400 MHz bandwidth signal **506** (e.g., shown as NB1 for RF chain **510-1**, NB2 for RF chain **510-2**, NB3 for RF chain **510-3**, and NB4 for RF chain **510-4**). Although FIG. **5**A illustrates the use of LPF, in some other examples, band pass filtering (BPF) may alternatively be applied. ADC converts the 100 MHz narrow band signal to a digital signal and FFT converts the time domain signal to the frequency domain. In an example, a maximum FFT size of 4K may be used to process the signal at each RF chain **510**. After each **100** MHz of the 400 MHz bandwidth signal **506** is processed by the RF chains **510**, phase alignment may be applied. Phase alignment may be optional, for example, phase alignment may be based on a capacity of UE **504** to perform phase alignment and/or based on a requirement for maintaining coherent phase in the wideband signal **506** (e.g., phase alignment may be necessary for certain applications, such as sensing applications). Further, concatenation may be applied to concatenate each 100 MHz signal processed by each RF chain **510** to create a wideband frequency domain (FD) signal (e.g., the 400 MHz signal **506**).

[0139] Some processing shown in FIG. **5**A for receiving a signal from network entity **502** may also be used to transmit a signal to network entity **502** over a contiguous frequency bandwidth. For example, in FIG. **5**B, to transmit a 400 MHz, uplink signal **508** to network entity **502**, UE **504** may use four RF chains **510-1**, **510-2**, **510-3**, and **510-4**, similar to FIG. **5**A, to process the signal **508** prior to transmission. The 400 MHz, uplink signal **508** may be segmented into four 100 MHz NB signals (e.g., (e.g., shown as NB1 for RF chain **510-1**, NB2 for RF chain **510-2**, NB3 for RF chain **510-3**, and NB4 for RF chain **510-4**), and each RF chain **510** may process one of the four 100 MHz signals by (1) performing IFFT (e.g., using a 4K FFT size) and (2) performing DAC to convert the **100** MHz signal to an analog signal.

[0140] Thus, as shown in FIGS. **5**A and **5**B, using multiple RF chains at UE **504** may enable UE **504** to process a wideband signal **506**, **508** without increasing the FFT size (e.g., increasing to an FFT size greater than 13K). Further, using multiple RF chains at UE **504** may enable UE **504** to process the wideband signal **506**, **508** without increasing overhead due to the use of guard bands or additional control signaling, which may occur when carrier aggregation is implemented, as described herein. Accordingly, multiple RF chains processing at a UE may allow for the use of larger bandwidth carriers to increase capacity of the wireless communications system.

[0141] While the use of multiple RF chains to communicate across contiguous frequency resources provides the aforementioned advantages, multiple RF chains processing at a UE may impact the UE's MIMO operation, when a MIMO operation is implemented for communicating with a network entity. In particular, a number of MIMO layers that may be used for MIMO communication (e.g., uplink and/or downlink communication) may be based on a number of RF chains available for communication at a transmitter and/or a receiver. For example, a UE capable of using four RF chains simultaneously for receiving and processing downlink communications from a network entity may be capable of receiving up to four MIMO layer transmissions in the downlink, from the network entity. However, when RF chain(s) at the UE are used to process a wideband signal, by each RF chain processing a subset of contiguous frequency resources of the wideband signal (e.g., each RF chain processing 100 MHz of a 400 MHz downlink signal), MIMO capability may decrease. In particular, a number of MIMO layers used for downlink communications with the UE may decrease. For example, each MIMO layer may require separate one or more RF chains to process the MIMO layer, where the one or more RF chains required to process a given MIMO layer is based on the carrier bandwidth (C_B) used for communication, and a maximum bandwidth (B_{MAX}) the UE can process per RF chain. For example, to process L MIMO layers, the UE may require R RF chains, where $R=L*C_B/B_{MAX}$. Accordingly, if R and B_{MAX} remain constant, as C_B increases, L decreases.

[0142] In a first illustrative example, a UE may include four RF chains (e.g., based on UE implementation), each capable of processing a maximum bandwidth equal to 100 MHz. Thus, when the UE receives a 400 MHz wideband signal, the UE may need to split the 400 MHz wideband signal into four narrowband 100 MHz signals to process the wideband signal (e.g., based on a capability of each RF chain at the UE to process a maximum bandwidth of 100 MHz). While a network entity may transmit up to four layers to the UE in a MIMO operation, in this case where all four RF chains of the UE are being used to process different subbands of the 400 MHz wideband signal, the network entity may use only one MIMO layer for downlink transmission when communicating with the UE.

[0143] In a second illustrative example, the same UE (e.g., capable of processing 100 MHz per RF chain and including four RF chains) may receive a 200 MHz signal instead of a 400 MHz signal. Based on an ability of each RF chain to process 100 MHz, the UE may need to split the 200 MHz signal into two narrowband 100 MHz signals for processing (e.g., 200 MHz signal/100 MHz processing capability per RF chain=2 narrowband signals). Each of the two narrowband signals may be processed by two of the RF chains at the UE (e.g., two RF chains process the first narrowband signal and two RF chains process the second RF signal). In this case, the network entity may use two MIMO layers for downlink transmission when communicating with the UE, but not more than two.

[0144] Due to the impact of multiple RF chain processing on MIMO operations, and more specifically on a number of MIMO layers that may be used for communication, as illustrated by the above two examples, a network entity may need to be informed about a UE's capability to use multiple RF chains at a time for communication.

[0145] Accordingly, aspects described herein provide techniques for reporting a UE's multiple RF chains processing capability to a network entity. For example, a UE may send, to a network entity, report(s) indicating a capability of the UE to use multiple RF chains at a single time to communicate across contiguous frequency resources, and more specifically, a capability of the UE to use each of the multiple RF chains to communicate across a subset of the contiguous frequency resources (e.g., a subband of a bandwidth used for communication). The network entity receiving the report(s) comprising such capability information for a UE may use this information when communicating with the UE. For example, the network entity may use this information when determining a number of MIMO layers to use when communicating with the UE.

[0146] In certain aspects, in addition to reporting a multiple RF chains processing capability of the UE, additional information may be reported to the network entity to help the network entity determine a number of MIMO layers to use for downlink and/or uplink communication with the UE. In certain aspects, this information includes an explicit indication of a maximum number of MIMO layers supported by the UE when using multiple RF chains at the UE according to the multiple RF chain processing capability of the UE. In certain aspects, this information implicitly indicates, to the network entity, a maximum number of MIMO layers supported by the UE when using multiple RF chains at the UE. For example, the network entity may use the reported information to determine a number of MIMO layers to use when communicating with the UE.

[0147] In certain aspects, different antenna elements (or arrays), associated with different RF chains at the UE, may not be co-located at the UE (e.g., these antenna elements/arrays/RF chains may be distributed around an enclosure of the UE). As a result, measurements for different reference signals received and processed by the UE using different RF chains at the UE may be different. For example, where the reference signals sent to the UE include channel state information RSs (CSI-RSs), different channel states may be observed by different RF chains of the UE. Thus, to obtain measurements over a contiguous frequency bandwidth (e.g., wideband), a UE may need to measure different subbands (e.g., different subsets of contiguous frequency resources associated with the wideband) with the same and/or different RF chains at the UE. Certain aspects described herein provide techniques for transmitting, to a UE, configurations of one or more RS occasions configured for communicating one or more RSs spanning the contiguous frequency bandwidth. With this configuration of RS occasions(s) over the bandwidth, a UE may be able to perform reference signal measurements using one or more RF chains at the UE. Accurate RS measurements, such as CSI-RS measurements, may be useful for scheduling and/or resource allocation for efficient communication between the UE and a network entity. Further, in certain aspects, RS measurements may be used by the UE to tune different RF chains at the UE to one or more subbands of a maximum bandwidth at the UE.

Example Operations of Entities in a Communications Network

[0148] FIG. **6** depicts a process flow **600** for communications in a network between a network entity **602** and a UE **604**. In certain aspects, the network entity **602** may be an example of the BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**. Similarly, the UE **604** may be an example of UE **104** depicted and described with respect to FIGS. **1** and **3**. However, in other aspects, UE **604** may be another type of wireless communications device and network entity **602** may be another type of network entity or network node, such as those described herein.

[0149] In certain aspects, process flow **600** is used to report, at least, a capability of UE **604** to use multiple RF chains at a time to communicate across contiguous frequency resources (e.g., a bandwidth, such as a 400 MHz carrier bandwidth, which may be supported in 6G and/or later generation wireless technologies). For example, as shown in FIG. **6**, at **608**, UE **604** sends one or more reports to network entity **602** indicating a multiple RF chains processing capability of UE **604**. The multiple RF chains processing capability of UE **604**, included in the report(s), may indicate, to network entity **602**, that UE **604** is capable of using multiple RF chains at a time to communicate across contiguous frequency resources, where each RF chain is used to communicate across a subset of the contiguous resources (e.g., each RF chain may be used to communicate across a subband of a larger bandwidth).

[0150] For example, as shown at **606**, UE **604** may be implemented to use four RF chains, where each RF chain is capable of processing a maximum bandwidth of 100 MHz. Accordingly, the indication to network entity **602**, included in report(s) sent from UE **604** at **608**, may indicate that UE **604** is capable of using multiple of the four RF chains at the UE for wideband communication (e.g., such as a 400 MHz communication), where each RF chain is used to communicate across a respective subband of the wideband communication.

[0151] In addition to reporting UE **604**'s multiple RF chains processing capability, in certain aspects, one or more other indications of capabilities of UE **604** are included in the one or more reports. Indications of other capabilities of UE **604** that may be reported to network entity **602** are shown in process flow **600** by transmissions **610-618**. Though shown as separate transmissions, the indications may be all sent in one transmission or report, or in any combination in any number of transmissions or reports.

[0152] For example, at **610**, UE **604** may send one or more reports to network entity **602** indicating a maximum FFT size per RF chain supported by UE **604**. At **612**, UE **604** may send one or more reports to network entity **602** indicating a maximum bandwidth per RF chain supported by UE **604**. At **614**, UE **604** may send one or more reports to network entity **602** indicating a maximum number of subsets of contiguous frequency resources (e.g., subbands) supported by UE **604**. In certain aspects, the number of subsets of contiguous frequency resources supported by UE **604** is equal to a maximum number of RF chains implemented at UE **604**. In certain other aspects, the number of subsets of contiguous frequency resources supported by UE **604** is less than a maximum number of RF chains implemented at UE **604**. For example, UE **604** may be implemented with three RF chains but be configured to use only two of the RF chains for wideband communication, where each of the two RF chains may be used to communicate across a different subband of a larger bandwidth (e.g., two subbands<three RF chains implemented at UE **604**).

[0153] In certain aspects (as described with respect to FIG. **7**A below), capability information sent to network entity **602** at **610**, **612**, **614**, and/or **616** may be used by network entity **602** to determine a maximum number of MIMO layers that may be used for uplink and/or downlink communications with UE **604**. As an illustrative example, UE **604** may send one or more reports, to network entity **602**, indicating (1) the multiple RF chains processing capability of UE **604** (e.g., at **610**), (2) a maximum FFT size per RF chain supported by UE **604** equal to 4K (4,096) (e.g., at **612**), and (3) a maximum number of subsets of contiguous frequency resources (e.g., subbands) supported by UE **604** equal to eight (e.g., at **614**). Network entity **602** may determine that eight RF chains may be used at UE **604** for wideband downlink communication based on the reported maximum number of subsets of contiguous frequency resources (e.g., subbands) supported by UE **604**. Further, network entity **602** may determine that UE **604** may receive a maximum of two layers in a downlink transmission if a 400 MHz bandwidth signal is transmitted with an SCS of 30 kHz. For example, as described herein, a single 400 MHz carrier may include a total of 13,104 tones. To process the 13,104 tones, network entity **602** may determine that UE **604** can use four of its eight RF chains, capable of processing a maximum FFT size of 4K, to process the 400 MHz signal. Thus, because only four of the eight RF chains may be needed to process the 400 MHz signal, network entity **602** may determine that a maximum of two MIMO layers may be used to communicate the **400** MHz signal to UE **604**.

[0154] In certain aspects, indications of other capabilities of UE **604** that may be reported to network entity **602** include an explicit indication of a maximum number of MIMO layers that are supported by UE **604** when using multiple RF chains to process communications from network entity **602**. For example, at **616**, UE **604** may send one or more reports to network entity **602** indicating a maximum number of MIMO layers supported by UE **604** when using multiple RF chains to process communications from network entity **602**. In some cases, UE **604** may send the indication at **616** as an alternative to indication(s) at **610**, **612**, and **614**, given the capability information reported at **610**, **612**, and **614** may be used by network entity **602** to derive the maximum number of MIMO layers, which is explicitly reported at **616**.

[0155] In certain aspects, indications of other capabilities of UE **604** that may be reported to network entity **602** include an indication of a capability of UE **604** to tune each RF chain at UE **604** to multiple subsets of a contiguous frequency band (e.g., tune each RF chain to multiple subbands of a contiguous frequency band). For example, at **618**, UE **604** may send one or more reports to network entity **602** indicating this capability. As described in detail below, network entity **602** may use this information to determine an RS configuration, with one or more RS occasions configured for communicating RSs, to send to UE **604**.

[0156] Although in process flow **600**, each indication is transmitted in a separate report to network entity **602** (e.g., at **612**, **614**, **616**, and **618**), in some other examples, one or more of these indications may be sent together and/or with the indication of UE **604**'s capability to use multiple RF chains communicated at **608**. Further, the order of different capability indications sent to network entity **602** may be different than the order illustrated in process flow **600**.

[0157] FIGS. **7**A and **7**B depict process flows **700***a*, **700***b* for communications in a network between a network entity **702** and a UE **704** to determine a number of MIMO layers (e.g., one or more layers up to four layers) to use for downlink and uplink communications, respectively. In certain aspects, the network entity **702** may be an example of the BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**. Similarly, the UE **704** may be an example of UE **104** depicted and described with respect to FIGS. **1** and **3**. However, in other aspects, UE **704** may be another type of wireless communications device and network entity **702** may be another type of network entity or network node, such as those described herein.

[0158] As shown in process flow **700***a* of FIG. **7**A, at **708**, network entity **702** determines a number of MIMO layers to use for downlink communications with UE **704** when UE **704** is using multiple RF chains for processing downlink communications. As described herein, network entity **702** may make this determination based on (1) an SCS configured for communication between network entity **702** and UE **704**, (2) a number of RF chains of UE **704** configured for communication between network entity **702** and UE **704**, (3) a maximum fast FFT size per RF chain supported by UE **704** (e.g., which may be communicated to network entity **702**, for example, as shown via communication **614** in FIG. **6**), (4) a maximum bandwidth per RF chain supported by supported by UE **704** (e.g., which may be communicated to network entity **702**, for example, as shown via communication **614** in FIG. **6**), and/or (5) a maximum number of subsets of contiguous frequency resources supported by supported by UE **704** (e.g., which may be communicated to network entity **702**, for example, as shown via communication **614** in FIG. **6**). In certain aspects, network entity **702** makes this determination further based on a maximum bandwidth, where the maximum bandwidth: (1) comprises a bandwidth configured for communication between network entity **702** and UE **704** or (2) is a nominal maximum system bandwidth of network entity **702**.

[0159] At **710** in process flow **700***a*, network entity **702** sends, to UE **704**, a downlink signal across a contiguous frequency bandwidth using the determined number of MIMO layers.

[0160] Different from process flow **700***a*, process flow **700***b* in FIG. **7**B is used to determine a number of MIMO layers to use for uplink communications with network entity **702** when UE **704** is using multiple RF chains for processing uplink communications for transmission. As shown in process flow **700***b* of FIG. **7**B, at **718**, UE **704** determines a number of MIMO layers to use for uplink communications with network entity **702** when UE **704** is using multiple RF chains for processing uplink communications. UE **704** may make this determination based on similar information used by network entity **702** when determining a number of MIMO layers to use for downlink communications at **708** in FIG. **7**A.

[0161] At **720** in process flow **700***b*, UE **704** sends, to network entity **702**, an uplink signal across a contiguous frequency bandwidth using the determined number of MIMO layers and multiple RF chains at UE **704**.

[0162] FIG. **8**A depicts a process flow **800** for communications in a network between a network entity **802** and a UE **804** to perform RS measurements. In certain aspects, the network entity **802** may be an example of the BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**. Similarly, the UE **804** may be an example of UE **104** depicted and described with respect to FIGS. **1** and **3**. However, in other aspects, UE **804** may be another type of wireless communications device and network entity **802** may be another type of network entity or network node, such as those described herein.

[0163] In certain aspects, process flow **800** is used to communicate, to UE **804**, a configuration of RS occasions configured for communicating RS(s) spanning a contiguous frequency bandwidth (e.g., 400 MHz). Based on the configuration, UE **804** may monitor the RS occasions for one or more RSs from network entity **802** to perform one or more RS measurements using the multiple RF chains at UE **804**. UE **804** may use the measurements to tune each RF chain of the plurality of RF chains to multiple subsets of the contiguous frequency bandwidth.

[0164] For example, as shown in process flow **800** of FIG. **8**A, at **810**, network entity **802** sends, to UE **804**, a configuration of one or more RS occasions configured for communicating RS(s) spanning a contiguous frequency bandwidth. RS occasions are time-frequency resources that UE **804** may monitor for RS(s) from network entity **802**.

[0165] In certain aspects, the RS occasions included in the configuration include multiple RS occasions that span different subsets of the contiguous frequency bandwidth. For example, as shown in FIG. **8**B, RS occasions configured at UE **804** may include RS occasions **852-1**, **852-2**, **852-3**, and **852-4** (collectively referred to herein as "RS occasions **852**" and individually referred to herein as "RS occasion **852**"). Each RS occasion **852** may span a subset of a total 400 MHz bandwidth, also referred to as a subband. For example, RS occasion **852-1** may span a first 100 MHz of the 400 MHz bandwidth, RS occasion **852-2** may span a second 100 MHz of the 400 MHz bandwidth, RS occasion **852-3** may span a third 100 MHz of the 400 MHz bandwidth, and RS occasion **852-4** may span a first four MHz of the 400 MHz bandwidth. RS occasions **852** may also occur at different times. For example, RS occasion **852-1** may occur earlier in time than RS occasion **852-2**, RS occasion **852-3**, and RS occasion **852-4**.

[0166] Although FIG. **8**B depicts only four RS occasions **852** associated with the configuration received at UE **804**, in certain other examples, more or less RS occasions may be associated with the configuration and thus configured at UE **804** for monitoring for RS(s).

[0167] In certain aspects, the RS occasions included in the configuration include RSs that span the entire contiguous frequency bandwidth. For example, as shown in FIG. **8**C, RS occasions configured at UE **804** may include RS occasions **854-1** through **854**-X (collectively referred to herein as "RS occasions **854**" and individually referred to herein as "RS occasion **854**") (e.g., where X is an integer greater than one). Each RS occasion **854** may span all contiguous frequency resources of a 400 MHz. In some cases, RS occasions **854-2** through **854**-X are RS occasions configured for re-communication of an RS communication in RS occasion **854-1**.

[0168] Returning to FIG. **8**A, UE **804** may monitor one or more RS occasions for one or more RSs based on the configuration received at **810**. For example, if the configuration received, by UE **804** at **810**, is the configuration of RS occasions **852** depicted in FIG. **8**B, UE **804** may monitor for, receive, and process RS(s) in different 100 MHz subbands over time. This monitoring configuration may allow UE **804** to receive and process RS(s) in each subband using each RF chain at UE **804**, assuming UE **804** has multiple RF chains with a maximum FFT size of 4K. This monitoring configuration, however, may not enable UE **804** to measure the entire 400 MHz at once using multiple RF chains at UE **804**.

[0169] Alternatively, if the configuration received, by UE **804** at **810**, is the configuration of RS occasion(s) **854** depicted in FIG. **8**C, UE **804** may monitor for, receive, and process an RS(s) transmitted across the entire 400 MHz bandwidth. However, UE **804** may need to use multiple RF chains at the UE to measure different subbands of the 400 MHz RS. Thus, with this configuration, less RS occasions may be configured and/or needed to generate a measurement for the 400 MHz bandwidth; however, not every RF chain at UE **804** may be used to measure every subband of the 400 MHz bandwidth.

[0170] Based on monitoring RS occasions **852** in FIG. **8**B and/or RS occasion(s) **854** in FIG. **8**C, at **814**, UE **804** may receive, process, and measure one or more RSs (e.g., RSs **812-1** through **812**-Z (collectively referred to herein as RSs **812**), where Z is an integer greater than one) using multiple RF chains at UE **804**. In certain aspects, the one or more RSs **812** are CSI-RS, and UE **804** measures each CSI-RS to determine channel state information measured by one or more RF chains for multiple subbands of the **400** MHz bandwidth.

[0171] Based on the measurement information, UE **804** may associate each RF chain with a different subband for processing. For example, a first RF chain at UE **804** may or may not be associated with a first subband of a configured bandwidth at UE **804** or a nominal maximum system bandwidth, a second RF chain at UE **804** may or may not be associated with a second subband of a configured bandwidth at UE **804** or a nominal maximum system bandwidth, etc. Instead, UE **804** may use the measurements to associate different RF chains with different subbands for processing. For example, an RF chain with a "best" RS measurement, such as a measurement for a CSI-RS indicating best channel conditions, for a subband, may be associated with that subband. As an illustrative example shown at **818** in FIG. **8**A, UE **804** may associate a third RF chain at UE **804** with a first subband **820-1**, associate a fourth RF chain at UE **804** with a second subband **820-2**, associate a second RF chain at UE **804** with a third subband **820-3**, and associated a first RF chain at UE **804** with a fourth subband **820-4**. Associating each RF chain of UE **804** with a subband of a configured frequency bandwidth may involve tuning each RF chain (e.g., each filter) of UE **804** to a subband of the configured frequency bandwidth. Although the example, in FIG. **8**A depicts each RF chain being tuned to a single subband, in some other examples, an RF chain may be tuned to multiple subbands.

Example Operations

[0172] FIG. **9** shows a method **900** for wireless communications by an apparatus, such as UE **104** of FIGS. **1** and **3**.

[0173] Method **900** begins at block **905** with transmitting, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources.

[0174] Method **900** then proceeds to block **910** with communicating, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth.

[0175] In certain aspects, the one or more reports further indicate a maximum FFT size per RF chain supported by the apparatus.

[0176] In certain aspects, the one or more reports further indicate a maximum bandwidth per RF chain supported by the apparatus.

[0177] In certain aspects, the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the apparatus.

[0178] In certain aspects, the one or more reports further indicate a maximum number of MIMO layers supported by the apparatus when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

[0179] In certain aspects, a maximum number of MIMO layers supported by the apparatus when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of: a maximum bandwidth; a SCS configured for communication between the network entity and the apparatus; a number of RF chains of the apparatus configured for communication between the network entity and the apparatus; a maximum FFT size per RF chain supported by the apparatus; a maximum bandwidth per RF chain supported by the apparatus; or a maximum number of subsets of contiguous frequency resources supported by the apparatus.

[0180] In certain aspects, the maximum bandwidth comprises a bandwidth configured for communication between the network entity and the apparatus.

[0181] In certain aspects, the maximum bandwidth comprises a nominal maximum system bandwidth of the network entity.

[0182] In certain aspects, method **900** further includes receiving a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

[0183] In certain aspects, each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

[0184] In certain aspects, each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

[0185] In certain aspects, the one or more reports further indicate a second capability of the apparatus to tune each RF chain of the plurality of RF chains to multiple subsets of the contiguous frequency bandwidth.

[0186] In certain aspects, the signal carries control information.

[0187] In certain aspects, block **910** includes: receiving the signal; or transmitting the signal.

[0188] In certain aspects, method **900**, or any aspect related to it, may be performed by an apparatus, such as communications device **1100** of FIG. **11**, which includes various components operable, configured, or adapted to perform the method **900**. Communications device **1100** is described below in further detail.

[0189] Note that FIG. **9** is just one example of a method, and other methods including fewer, additional, or alternative operations are possible consistent with this disclosure.

[0190] FIG. **10** shows a method **1000** for wireless communications by an apparatus, such as BS **102** of FIGS. **1** and **3**, or a disaggregated base station as discussed with respect to FIG. **2**.

[0191] Method **1000** begins at block **1005** with receiving, from a UE, one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources.

[0192] Method **1000** then proceeds to block **1010** with communicating, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth.

[0193] In certain aspects, the one or more reports further indicate a maximum FFT size per RF chain supported by the UE.

[0194] In certain aspects, the one or more reports further indicate a maximum bandwidth per RF chain supported by the UE.

[0195] In certain aspects, the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the UE.

[0196] In certain aspects, the one or more reports further indicate a maximum number of MIMO layers supported by the UE when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

[0197] In certain aspects, method **1000** further includes determining a maximum number of MIMO layers supported by the UE when the UE is operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of: a maximum bandwidth; a SCS configured for communication between the apparatus and the UE; a number of RF chains of the UE configured for communication between the apparatus and the UE; a maximum FFT size per RF chain supported by the UE; a maximum bandwidth per RF chain supported by the UE; or a maximum number of subsets of contiguous frequency resources supported by the UE.

[0198] In certain aspects, the maximum bandwidth comprises a bandwidth configured for communication between the UE and the apparatus.

[0199] In certain aspects, the maximum bandwidth comprises a nominal maximum system bandwidth of the apparatus.

[0200] In certain aspects, method **1000** further includes transmitting a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

[0201] In certain aspects, each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

[0202] In certain aspects, each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

[0203] In certain aspects, the one or more reports further indicate a second capability of the UE to tune each RF chain of a plurality of RF chains of the UE to multiple subsets of the contiguous frequency bandwidth.

[0204] In certain aspects, the signal carries control information.

[0205] In certain aspects, block **1010** includes: transmitting the signal; or receiving the signal.

[0206] In certain aspects, method **1000**, or any aspect related to it, may be performed by an apparatus, such as communications device **1200** of FIG. **12**, which includes various components operable, configured, or adapted to perform the method **1000**. Communications device **1200** is described below in further detail.

[0207] Note that FIG. **10** is just one example of a method, and other methods including fewer, additional, or alternative operations are possible consistent with this disclosure.

Example Communications Devices

[0208] FIG. **11** depicts aspects of an example communications device **1100**. In some aspects, communications device **1100** is a user equipment, such as UE **104** described above with respect to FIGS. **1** and **3**.

[0209] The communications device **1100** includes a processing system **1105** coupled to a transceiver **1155** (e.g., a transmitter and/or a receiver). The transceiver **1155** is configured to transmit and receive signals for the communications device **1100** via an antenna **1160**, such as the various signals as described herein. The processing system **1105** may be configured to perform processing functions for the communications device **1100**, including processing signals received and/or to be transmitted by the communications device **1100**.

[0210] The processing system **1105** includes one or more processors **1110**. In various aspects, the one or more processors **1110** may be representative of one or more of receive processor **358**, transmit processor **364**, TX MIMO processor **366**, and/or controller/processor **380**, as described with respect to FIG. **3**. The one or more processors **1110** are coupled to a computer-readable medium/memory **1130** via a bus **1150**. In certain aspects, the computer-readable medium/memory **1130** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1110**, enable and cause the one or more processors **1110** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it, including any operations described in relation to FIG. **9**. Note that reference to a processor performing a function of communications device **1100** may include one or more processors performing that function of communications device **1100**, such as in a distributed fashion.

[0211] In the depicted example, computer-readable medium/memory **1130** stores code for sending **1135**, code for communicating **1140**, and code for receiving **1145**. Processing of the code **1135-1145** may enable and cause the communications device **1100** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it.

[0212] The one or more processors **1110** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1130**, including circuitry for sending **1115**, circuitry for communicating **1120**, and circuitry for receiving **1125**. Processing with circuitry **1115-1125** may enable and cause the communications device **1100** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it.

[0213] More generally, means for communicating, transmitting, sending or outputting for transmission may include the transceivers **354**, antenna(s) **352**, transmit processor **364**, TX MIMO processor **366**, AI processor **370**, and/or controller/processor **380** of the UE **104** illustrated in FIG. **3**, transceiver **1155** and/or antenna **1160** of the communications device **1100** in FIG. **11**, and/or one or more processors **1110** of the communications device **1100** in FIG. **11**. Means for communicating, receiving or obtaining may include the transceivers **354**, antenna(s) **352**, receive processor **358**, AI processor **370**, and/or controller/processor **380** of the UE **104** illustrated in FIG. **3**, transceiver **1155** and/or antenna

1160 of the communications device **1100** in FIG. **11**, and/or one or more processors **1110** of the communications device **1100** in FIG. **11**.

[0214] FIG. **12** depicts aspects of an example communications device **1200**. In some aspects, communications device **1200** is a network entity, such as BS **102** of FIGS. **1** and **3**, or a disaggregated base station as discussed with respect to FIG. **2**.

[0215] The communications device **1200** includes a processing system **1205** coupled to a transceiver **1265** (e.g., a transmitter and/or a receiver) and/or a network interface **1275**. The transceiver **1265** is configured to transmit and receive signals for the communications device **1200** via an antenna **1270**, such as the various signals as described herein. The network interface **1275** is configured to obtain and send signals for the communications device **1200** via communications link(s), such as a backhaul link, midhaul link, and/or fronthaul link as described herein, such as with respect to FIG. **2**. The processing system **1205** may be configured to perform processing functions for the communications device **1200**, including processing signals received and/or to be transmitted by the communications device **1200**.

[0216] The processing system **1205** includes one or more processors **1210**. In various aspects, one or more processors **1210** may be representative of one or more of receive processor **338**, transmit processor **320**, TX MIMO processor **330**, and/or controller/processor **340**, as described with respect to FIG. **3**. The one or more processors **1210** are coupled to a computer-readable medium/memory **1235** via a bus **1260**. In certain aspects, the computer-readable medium/memory **1235** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1210**, enable and cause the one or more processors **1210** to perform the method **1000** described with respect to FIG. **10**, or any aspect related to it, including any operations described in relation to FIG. **10**. Note that reference to a processor of communications device **1200** performing a function may include one or more processors of communications device **1200** performing that function, such as in a distributed fashion.

[0217] In the depicted example, the computer-readable medium/memory **1235** stores code for receiving **1240**, code for communicating **1245**, code for determining **1250**, and code for transmitting **1255**. Processing of the code **1240-1255** may enable and cause the communications device **1200** to perform the method **1000** described with respect to FIG. **10**, or any aspect related to it.

[0218] The one or more processors **1210** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1235**, including circuitry for receiving **1215**, circuitry for communicating **1220**, circuitry for determining **1225**, and circuitry for transmitting **1230**. Processing with circuitry **1215-1230** may enable and cause the communications device **1200** to perform the method **1000** described with respect to FIG. **10**, or any aspect related to it.

[0219] More generally, means for communicating, transmitting, sending or outputting for transmission may include the transceivers **332**, antenna(s) **334**, transmit processor **320**, TX MIMO processor **330**, AI processor **318**, and/or controller/processor **340** of the BS **102** illustrated in FIG. **3**, transceiver **1265**, antenna **1270**, and/or network interface **1275** of the communications device **1200** in FIG. **12**, and/or one or more processors **1210** of the communications device **1200** in FIG. **12**. Means for communicating, receiving or obtaining may include the transceivers **332**, antenna(s) **334**, receive processor **338**, AI processor **318**, and/or controller/processor **340** of the BS **102** illustrated in FIG. **3**, transceiver **1265**, antenna **1270**, and/or network interface **1275** of the communications device **1200** in FIG. **12**, and/or one or more processors **1210** of the communications device **1200** in FIG. **12**.

Example Clauses

[0220] Implementation examples are described in the following numbered clauses:

[0221] Clause 1: A method for wireless communications by an apparatus comprising: transmitting, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and communicating, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth.

[0222] Clause 2: The method of Clause 1, wherein the one or more reports further indicate a maximum FFT size per RF chain supported by the apparatus.

[0223] Clause 3: The method of any one of Clauses 1-2, wherein the one or more reports further indicate a maximum bandwidth per RF chain supported by the apparatus.

[0224] Clause 4: The method of any one of Clauses 1-3, wherein the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the apparatus.

[0225] Clause 5: The method of any one of Clauses 1-4, wherein the one or more reports further indicate a maximum number of MIMO layers supported by the apparatus when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

[0226] Clause 6: The method of any one of Clauses 1-5, wherein a maximum number of MIMO layers supported by the apparatus when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of: a maximum bandwidth; a SCS configured for communication between the network entity and the apparatus; a number of RF chains of the apparatus configured for communication between the network entity and the apparatus; a maximum FFT size per RF chain supported by the apparatus; a maximum bandwidth per RF chain supported by the apparatus; or a maximum number of subsets of contiguous frequency resources supported by the apparatus.

[0227] Clause 7: The method of Clause 6, wherein the maximum bandwidth comprises a bandwidth configured for communication between the network entity and the apparatus.

[0228] Clause 8: The method of Clause 6, wherein the maximum bandwidth comprises a nominal maximum system bandwidth of the network entity.

[0229] Clause 9: The method of any one of Clauses 1-8, further comprising: receiving a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

[0230] Clause 10: The method of Clause 9, wherein each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

[0231] Clause 11: The method of Clause 9, wherein each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

[0232] Clause 12: The method of any one of Clauses 1-11, wherein the one or more reports further indicate a second capability of the apparatus to tune each RF chain of the plurality of RF chains to multiple subsets of the contiguous frequency bandwidth.

[0233] Clause 13: The method of any one of Clauses 1-12, wherein the signal carries control information.

[0234] Clause 14: The method of any one of Clauses 1-13, wherein communicating the signal comprises: receiving the signal; or transmitting the signal.

[0235] Clause 15: A method for wireless communications by an apparatus comprising: receiving, from a UE, one or more reports indicating a first capability of the UE to use multiple RF chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and communicating, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth.

[0236] Clause 16: The method of Clause 15, wherein the one or more reports further indicate a maximum FFT size per RF chain supported by the UE.

[0237] Clause 17: The method of any one of Clauses 15-16, wherein the one or more reports further indicate a maximum bandwidth per RF chain supported by the UE.

[0238] Clause 18: The method of any one of Clauses 15-17, wherein the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the UE.

[0239] Clause 19: The method of any one of Clauses 15-18, wherein the one or more reports further indicate a maximum number of MIMO layers supported by the UE when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

[0240] Clause 20: The method of any one of Clauses 15-19, further comprising determining a maximum number of MIMO layers supported by the UE when the UE is operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of: a maximum bandwidth; a SCS configured for communication between the apparatus and the UE; a number of RF chains of the UE configured for communication between the apparatus and the UE; a maximum FFT size per RF chain supported by the UE; a maximum bandwidth per RF chain supported by the UE; or a maximum number of subsets of contiguous frequency resources supported by the UE.

[0241] Clause 21: The method of Clause 20, wherein the maximum bandwidth comprises a bandwidth configured for communication between the UE and the apparatus.

[0242] Clause 22: The method of Clause 20, wherein the maximum bandwidth comprises a nominal maximum system bandwidth of the apparatus.

[0243] Clause 23: The method of any one of Clauses 15-22, further comprising: transmitting a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

[0244] Clause 24: The method of Clause 23, wherein each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

[0245] Clause 25: The method of Clause 23, wherein each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

[0246] Clause 26: The method of any one of Clauses 15-25, wherein the one or more reports further indicate a second capability of the UE to tune each RF chain of a plurality of RF chains of the UE to multiple subsets of the contiguous frequency bandwidth.

[0247] Clause 27: The method of any one of Clauses 15-26, wherein the signal carries control information.

[0248] Clause 28: The method of any one of Clauses 15-27, wherein communicating the signal comprises: transmitting the signal; or receiving the signal.

[0249] Clause 29: One or more apparatuses, comprising: one or more memories comprising executable instructions; and one or more processors configured to execute the executable instructions and cause the one or more apparatuses to perform a method in accordance with any one of Clauses 1-28.

[0250] Clause 30: One or more apparatuses, comprising: one or more memories; and one or more processors, coupled to the one or more memories, configured to cause the one or more apparatuses to perform a method in accordance with any one of Clauses 1-28.

[0251] Clause 31: One or more apparatuses, comprising: one or more memories; and one or more processors, coupled to the one or more memories, configured to perform a method in accordance with any one of Clauses 1-28.

[0252] Clause 32: One or more apparatuses, comprising means for performing a method in accordance with any one of Clauses 1-28.

[0253] Clause 33: One or more non-transitory computer-readable media comprising executable instructions that, when executed by one or more processors of one or more apparatuses, cause the one or more apparatuses to perform a method in accordance with any one of Clauses 1-28.

[0254] Clause 34: One or more computer program products embodied on one or more computer-readable storage media comprising code for performing a method in accordance with any one of Clauses 1-28.

[0255] Clause 35: A user equipment (UE), comprising: a processing system that includes processor circuitry and memory circuitry that stores code and is coupled with the processor circuitry, the processing system configured to cause the UE to perform a method in accordance with any one of Clauses 1-14.

[0256] Clause 36: A network entity, comprising: a processing system that includes processor circuitry and memory circuitry that stores code and is coupled with the processor circuitry, the processing system configured to cause the network entity to perform a method in accordance with any one of Clauses 15-28.

Additional Considerations

[0257] The preceding description is provided to enable any person skilled in the art to practice the various aspects described herein. The examples discussed herein are not limiting of the scope, applicability, or aspects set forth in the claims. Various modifications to these aspects will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other aspects. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various actions may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

[0258] The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a general purpose processor, an AI processor, a digital signal processor (DSP), an ASIC, a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any commercially available processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, a system on a chip (SoC), or any other such configuration.

[0259] As used herein, a phrase referring to "at least one of" a list of items refers to any combination of those items, including single members. As an example, "at least one of: a, b, or c" is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

[0260] As used herein, the term "determining" encompasses a wide variety of actions. For example, "determining" may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, "determining" may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, "determining" may include resolving, selecting, choosing, establishing and the like.

[0261] As used herein, "coupled to" and "coupled with" generally encompass direct coupling and indirect coupling (e.g., including intermediary coupled aspects) unless stated otherwise. For example, stating that a processor is coupled to a memory allows for a direct coupling or a coupling via an intermediary aspect, such as a bus.

[0262] The methods disclosed herein comprise one or more actions for achieving the methods. The method actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of actions is specified, the order and/or use of specific actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor.

[0263] The following claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims. Reference to an element in the singular is not intended to mean only one unless specifically so stated, but rather "one or more." The subsequent use of a definite article (e.g., "the" or "said") with an element (e.g., "the processor") is not intended to invoke a singular meaning (e.g., "only one") on the element unless otherwise specifically stated. For example, reference to an element (e.g., "a processor," "a controller," "a memory," "a transceiver," "an antenna," "the processor," "the controller," "the memory," "the transceiver," "the antenna," etc.), unless otherwise specifically stated, should be understood to refer to one or more elements (e.g., "one or more processors," ""one" or more controllers," "one or more memories," "one more transceivers," etc.). The terms "set" and "group" are intended to include one or more elements, and may be used interchangeably with "one or more." Where reference is made to one or more elements performing functions (e.g., steps of a method), one element may perform all functions, or more than one element may collectively perform the functions. When more than one element collectively performs the functions, each function need not be performed by each of those elements (e.g., different functions may be performed by different elements) and/or each function need not be performed in whole by only one element (e.g., different elements may perform different sub-functions of a function). Similarly, where reference is made to one or more elements configured to cause another element (e.g., an apparatus) to perform functions, one element may be configured to cause the other element to perform all functions, or more than one element may collectively be configured to cause the other element to perform the functions. Unless specifically stated otherwise, the term "some" refers to one or more. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

What is claimed is:

1. An apparatus configured for wireless communications, comprising:

one or more memories; and

one or more processors coupled to the one or more memories, the one or more processors being configured to cause the apparatus to:

transmit, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple radio frequency (RF) chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and

communicate, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth.

2. The apparatus of claim **1**, wherein the one or more reports further indicate a maximum fast Fourier transform (FFT) size per RF chain supported by the apparatus.

3. The apparatus of claim **1**, wherein the one or more reports further indicate a maximum bandwidth per RF chain supported by the apparatus.

4. The apparatus of claim **1**, wherein the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the apparatus.

5. The apparatus of claim **1**, wherein the one or more reports further indicate a maximum number of multi-input-multiple-output (MIMO) layers supported by the apparatus when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

6. The apparatus of claim **1**, wherein a maximum number of multi-input-multiple-output (MIMO) layers supported by the apparatus when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of:

a maximum bandwidth;

a subcarrier spacing (SCS) configured for communication between the network entity and the apparatus;

a number of RF chains of the apparatus configured for communication between the network entity and the apparatus;

a maximum fast Fourier transform (FFT) size per RF chain supported by the apparatus;

a maximum bandwidth per RF chain supported by the apparatus; or

a maximum number of subsets of contiguous frequency resources supported by the apparatus.

7. The apparatus of claim **6**, wherein the maximum bandwidth comprises a bandwidth configured for communication between the network entity and the apparatus.

8. The apparatus of claim **6**, wherein the maximum bandwidth comprises a nominal maximum system bandwidth of the network entity.

9. The apparatus of claim **1**, wherein the one or more processors are configured to cause the apparatus to:

receive a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

10. The apparatus of claim **9**, wherein each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

11. The apparatus of claim **9**, wherein each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

12. The apparatus of claim **1**, wherein the one or more reports further indicate a second capability of the apparatus to tune each RF chain of the plurality of RF chains to multiple subsets of the contiguous frequency bandwidth.

13. The apparatus of claim **1**, wherein the signal carries control information.

14. The apparatus of claim **1**, wherein to communicate the signal, the one or more processors are configured to:

receive the signal; or

transmit the signal.

15. An apparatus configured for wireless communications, comprising:

one or more memories; and

one or more processors coupled to the one or more memories, the one or more processors being configured to cause the apparatus to:

receive, from a user equipment (UE), one or more reports indicating a first capability of the UE to use multiple radio frequency (RF) chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and

communicate, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth.

16. The apparatus of claim **15**, wherein the one or more reports further indicate a maximum fast Fourier transform (FFT) size per RF chain supported by the UE.

17. The apparatus of claim **15**, wherein the one or more reports further indicate a maximum bandwidth per RF chain supported by the UE.

18. The apparatus of claim **15**, wherein the one or more reports further indicate a maximum number of subsets of contiguous frequency resources supported by the UE.

19. The apparatus of claim **15**, wherein the one or more reports further indicate a maximum number of multi-input-multiple-output (MIMO) layers supported by the UE when operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources.

20. The apparatus of claim **15**, wherein the one or more processors are configured to cause the apparatus to determine a maximum number of multi-input-multiple-output (MIMO) layers supported by the UE when the UE is operating according to the first capability of using multiple RF chains to communicate across contiguous frequency resources is based on at least one of:

a maximum bandwidth;

a subcarrier spacing (SCS) configured for communication between the apparatus and the UE;

a number of RF chains of the UE configured for communication between the apparatus and the UE;

a maximum fast Fourier transform (FFT) size per RF chain supported by the UE;

a maximum bandwidth per RF chain supported by the UE; or

a maximum number of subsets of contiguous frequency resources supported by the UE.

21. The apparatus of claim **20**, wherein the maximum bandwidth comprises a bandwidth configured for communication between the UE and the apparatus.

22. The apparatus of claim **20**, wherein the maximum bandwidth comprises a nominal maximum system bandwidth of the apparatus.

23. The apparatus of claim **15**, wherein the one or more processors are configured to cause the apparatus to:

transmit a configuration of one or more reference signal occasions configured for communicating one or more reference signals spanning the contiguous frequency bandwidth.

24. The apparatus of claim **23**, wherein each of the one or more reference signal occasions spans the contiguous frequency bandwidth.

25. The apparatus of claim **23**, wherein each of the one or more reference signal occasions spans a respective subset of the contiguous frequency bandwidth.

26. The apparatus of claim **15**, wherein the one or more reports further indicate a second capability of the UE to tune each RF chain of a plurality of RF chains of the UE to multiple subsets of the contiguous frequency bandwidth.

27. The apparatus of claim **15**, wherein the signal carries control information.

28. The apparatus of claim **15**, wherein to communicate the signal, the one or more processors are configured to:

transmit the signal; or

receive the signal.

29. A method for wireless communications by an apparatus comprising:

sending, to a network entity, one or more reports indicating a first capability of the apparatus to use multiple radio frequency (RF) chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources; and

communicating, with the network entity, a signal across a contiguous frequency bandwidth, wherein each of a plurality of RF chains of the apparatus is used to communicate the signal across a respective subset of the contiguous frequency bandwidth.

30. A method for wireless communications by an apparatus comprising:

receiving, from a user equipment (UE), one or more reports indicating a first capability of the UE to use multiple radio frequency (RF) chains at a time to communicate across contiguous frequency resources and to use each of the multiple RF chains to communicate across a respective subset of the contiguous frequency resources consisting of less than all of the contiguous frequency resources; and

communicating, with the UE, based on the first capability, a signal across a contiguous frequency bandwidth.

* * * * *